TABLE OF HISTORY OF CHANGES	
Version (date)	Changes
Version 1 (22/08/2022)	Annex 1 Part A
	Addition of deliverable updates as new deliverables: • D1.4 RRI First Progress Report for M30 • D1.5 RRI Second Progress Report for M48 • D8.4 Dissemination and Communication Plan (first update) for M20 • D8.5 Dissemination and Communication Plan (first update) for M36 • D8.6 REDOL Exploitation strategy and IPR manual (first update) for M30 • D8.7 REDOL Exploitation strategy and IPR manual (second update) for M48 Modification of deliverable names: • D1.3 First Report and Progress Report by D1.3 RRI First Report Addition of 2 new Critical Risks
Version 1 (22/08/2022)	Annex 1 Part B
	Clarifications requested through GAP document added: • Chapter 1.1.2 Ambition • Chapter 1.2.1 Overall concept and approach: Aragon as a Hub for Circularity • Chapter 2.1.1 Contribution of the project to the expected outcomes specified in the topic • Chapter 2.2.2 Exploitation of results • Chapter 3.2 Capacity of participants and consortium as a whole Addition of numbering of titles: • 3.1.g. 'Subcontracting' costs items • 3.1.h. 'Purchase costs' items Breakdown of costs added. • 3.1.h. "Purchase costs" items table.
Version 2 (10/10/2022)	Annex 1 Part A
	Modification of budget • Beneficiary 17/suite5 has changed its budget including unit costs for SME owner not receiving a salary • Beneficiary 3/AITEX has changed its budget including a subcontracting cost (characterisation analysis within T4.4) and a new OG&S cost (reactants and consumables for polymerization process). The total budget remains equal. • Beneficiary 4/ICSS has changed its budget switching a cost from Equipment to OG&S (Consumables 3 x CPU and electronics (3x11k€) (All in WP3)). The total budget remains equal. • Beneficiary 24/SAI has been deleted from the project and CIRCE has assumed its budget. Modification of DoA: • Beneficiary 1/CIRCE has been included as involved partner in the heading of T4.4 (typo error) • Extend T2.6 until M15 in order to have more data from partners to define very well the KPIs. • Beneficiary 6/ITA: move 9 PMs from WP6 to WP2 (T2.4) to start working as soon as possible with FINSA on incorporating fibreglass into the wood panels. 27/FINSA already has 6 PMs in WP2 but 6/ITA have none. 6 PMs remain in WP6 for the final optimisation and demonstration part. • Beneficiary 22/ACT has increased its efforts in WP4 and WP6 in 2 PMs each.
Version 2 (10/10/2022)	Annex 1 Part B
	Breakdown of costs added. • 3.1.h. "Purchase costs" items table has been modified following the EC's requests • Modifications of Beneficiary 3/AITEX, 4/ICSS, 22/ACT (described above) added • Beneficiary 14/IRIS has included a description of an item (IT components) within OG&S category Update of Gantt

	• Extend T2.6 until M15 in order to have more data from partners to define very well the KPIs.
Version 3 (09/02/2023)	Annex 1 Part A
	Legal status: • Beneficiary 19/STRA has changed his legal status. It has implied a modification of the budget items although the total budget remains equal. Efforts has changed from 89PMs to 50PMs due to a readjustment and also a typo error although the scope remains equal. Modifications in budget: • Beneficiary 3/AITEX has included an equipment with a cost of 51.5000€ for the amortization of a 50 L reactor connected to a continuous filtration system and has reduced the PM cost from 4.500€/PM to 3.848,10 €/PM after doing the actual distribution of the people per WP and noted that the average personnel cost is lower than initially estimated. The total budget remains equal. • Due to a mistake in the proposal phase, the development of the website was assigned to Beneficiary 1/CIRCE but the budget for it to Beneficiary 20/GEO. Therefore, GEO has transfered 8.000€ + 25% indirects (total 10.000€) already included in their budget for the creation of website to CIRCE, that implies 4800€ + 25% indirects (total 6000€ because GEO has 60% of financing). This allows CIRCE to develop the expected website using their current efforts. Modifications in DoA: • Beneficiary 6/ITA moves 2PM from WP8 to WP4 • Beneficiary 7/UZ increases 24PMs their efforts (11PMs in WP1, +8PMs in WP2, +3 PMs in WP4, +8PMs in WP7 and +8PMs in WP8). They reduce the average cost from 4.000€/PM to 3.024,67€/PM and the total budget remains equal. • Included beneficiary 6/ITA as participant in T2.4 (typo error). • Changed due date of D2.3 from M12 to M15. This deliverable is associated to T2.6 that finishes in M15 • Changed due date of MS5 from M24 to M36 since it is linked to the release of D6.1 • Changed due date of MS7 from M18 to M48 since it is linked to the release of D5.3
Version 3 (09/02/2023	Annex 1 Part B
	Change Table 9, specifically owner of KER10 from ITE to NTT due to a typo error Update values of Table 11 List of work packages Update 3.1.h. “Purchase costs” items table after modification of Beneficiary 1/CIRCE and 20/GEO within OG&S category Update of Gantt (typo errors) • Stablish RINA-C CSM as leader of T5.4 • Stablish CIRCE as leader of T8.3

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EXECUTIVE SUMMARY

Problem and significance

Circular economy (CE) concepts, endorsed at European level through the **CE Action Plan (COM/2020/98)**, aim to achieve improved waste prevention and resource management by extending and closing material cycles. **Solid urban waste (SUW)** is an abundant source to produce circular products, but it is generally not exploited. In fact, over **500 kg of municipal waste per capita** were generated in the EU in 2020, while **only 45% was recycled**. The proximity of resources and people, a sufficient scale for effective markets and the ability to shape urban planning and policy are key factors for **cities to achieve advancements in this area, fostering circular economy strategies** and **industrial-urban symbiosis (I-US) approaches** among local and regional actors. Through this approach, cities must become a cornerstone of the Circular Economy Package to achieve the **2030 targets for recycling** (65% for SUW, 70% for Construction demolition waste (CDW) and 75% for packaging). However, there is a need to overcome major existing barriers first, for which it is necessary to place the focus on the entire SUW value chains and promote business models that create an environment in which all parties involved perceive a win-win situation.

Solution

REDOL will redesign **five value chains for SUW** in **Aragon region** to produce circular products through an **I-US approach**. To this end, REDOL will overcome the identified barriers at EU and local level by developing and displaying a range of new solutions for; (i) **upgrading management technologies** to collect, sort and classify SUW, (ii) **enhancing the processing techniques** (e.g., chemical recycling) of sorted materials to avoid landfilling and (iii) **applying cutting-edge digital tools** to optimise recycling routes and interaction among key players. All these elements will be **validated in operational environments** coming from local waste managers and involving industrial facilities from the following value chains: **packaging, plastics, CDW**, **textiles and waste electric and electronic equipment** (WEEE). These solutions will result in the production of circular products such as civil engineering structures, geotextiles, reinforced wood-boards, packaging, etc. Moreover, REDOL will provide the required **organisational procedures, business models and social innovation actions** required for the establishment of successful symbiotic interactions and hubs for circularity at local level. Such an approach will result in the development of **guidelines and recommendations for major decision-making bodies and initiatives** at local and EU level and will achieve **improved citizens' perception** on SUW as a local resource. In this way, REDOL will achieve **consumer acceptance** of recycled products, thus increasing their participation in separate collection schemes at municipality level.

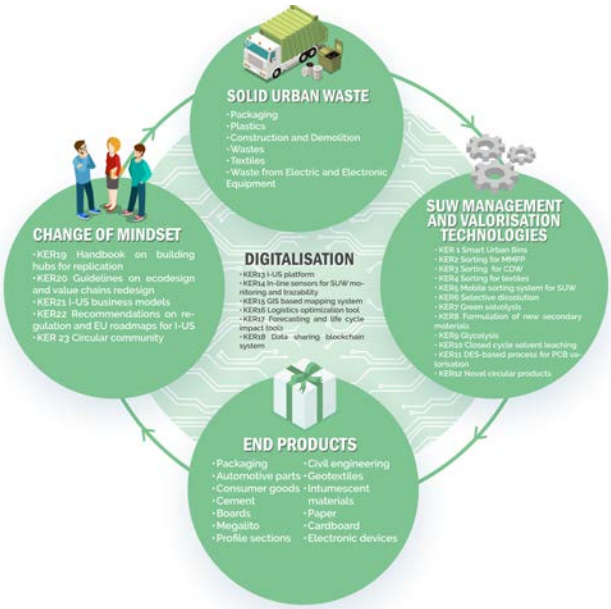

Figure 1. REDOL´s approach: Closing the Loop

Team

REDOL addresses this complex challenge by gathering a competitive consortium of 35 partners, from 12 EU countries, counting with CIRCE as coordinator and **I-US facilitator at local level**, and Zaragoza city council as **main public body occupying the central hierarchical position of the hub for circularity**. The nexus with local industry and stakeholders is performed through **5 public and private waste managers and a total of 8 industrial facilities/end-users**. In addition, the involvement of 3 cities will enable the **initiation of REDOL replication strategy**. The consortium complementarity is based on the diversified expertise, knowledge and networks, substantially **covering all addressed SUW value chains**, skills, resources and access to stakeholders. Additionally, several REDOL partners are active in **the Process4planet partnership** and the funded **ECoP on Hubs for Circularity.** Accordingly, a bi-directional flow of information is ensured for **increasing the exchange of knowledge at EU level and maximizing the impact of the project results**.

Impact

REDOL will directly impact in the **transition of Zaragoza towards a zero residues city**, aiming at achieving this objective by 2040. Such an approach will result in a total of **144.720 tons/year being re-used, valorised or transformed into secondary raw materials instead of landfilled** (over 90.000 tons in the value chains addressed in REDOL). Accordingly, REDOL innovations at technical, economical, managerial and social level will enable the consolidation of Zaragoza's hub for circularity and **leverage benefits within the city of over 14 B€ and 280 ktCO_2/year, economic and GHG emissions savings** respectively.

1. EXCELLENCE

1.1. Objectives and ambition

1.1.1. Objectives

The overall objective of REDOL is to **advance the technological, managerial, economic and social readiness level of EU Hubs for circularity** by demonstrating innovative and sustainable routes to **valorise SUW flows through I-US approaches**. To this end, five different value chains for **packaging, plastics, CDW, textiles and WEEE** will be developed and implemented in **real waste management facilities**, obtaining recycled secondary materials for the latter production of **high added value circular products**.

The **specific goals** that will enable the achievement of REDOL main objective are presented below:

Specific Objective 1: To develop and demonstrate **novel management systems and processing technologies** for the valorisation of SUW to be used as feedstock for other sectors and/or across value chains.

REDOL will demonstrate different valorisation routes for packaging, plastics, CDW, textiles and WEEE streams. The processes will be focussed on SUW that are currently disposed of, thus reducing the ratio of landfilling and incineration. The developed solutions, including collection, sorting, classification, processing and recycling technologies, will be implemented through a demonstration campaign treating real waste streams coming from local waste managers and industries, validating the technologies' performance in real operation conditions.

Specific:	Linked Deliverables & KER: D3.1, D3.2, D4.1, D4.2, KER1-11
Measurable:	Linked Impact Factor (IF): IF1, IF5 (see section 2.1.3)
Achievable:	Linked Partners: CIRCE, AIT, AITEX, ICCS, ITE, NTT, VDZ, IRIS, BRI, RINA, AMB
Relevant:	Relation to the topic (scope's bullet points): #1"Management and processing of waste", #4"Integration of novel sensing technology, IoT and digital tools"
Time-bound:	Linked work packages and tasks: T2.4, WP3, WP4, WP6

Specific Objective 2: To **obtain recycled secondary materials** from SUW ready to be fed in downstream processes for the **further production of circular products**, closing the production loop.

As a result of the SUW valorisation processes, REDOL will produce secondary materials that can be used as feedstocks in local industries to produce new circular products (packaging, construction components, wood boards, etc.). The quality of circular products will be validated by industrial companies from the corresponding sectors, thus ensuring the straightforward integration of these products in downstream applications. In order to overcome barriers for their entrance to market, recommendations on regulations and eco-design practices will be provided, and end-users demanding these materials will be involved within REDOL Circular Community, putting in place the agreements and business models required for their final commercialization.

Specific:	Linked Deliverables & KER: D5.3, D6.2, KER12, KER23
Measurable:	Linked Impact Factor (IF): IF2, IF5 (see section 2.1.3)
Achievable:	Linked Partners: ITA, TEC, ACC, CEM, FIN
Relevant:	Relation to the topic (scope's bullet points): #1"Valorisation of waste to be used as feedstock for other plants and companies across sectors and/or across value chains", #5"New approach to end-of life materials"
Time-bound:	Linked work packages and tasks: WP4, WP6

Specific Objective 3: To enable the achievement of a **zero SUW landfilling** approach in Zaragoza by accomplishing **55-100% recycling rates** in REDOL value chains while **creating new sustainable economic activity** in the Aragon region.

REDOL will reduce the overall environmental impact related to the management/disposal of SUW by increasing the recovery rate of untapped or discarded waste streams while enhancing the energy efficiency and sustainability of the processes. To integrate REDOL development in an optimised way, the project will develop an I-US Platform, that will provide a common framework for a standardised and secure data gathering and sharing, and for the monitoring and management of the value chains in a harmonised manner. This will also ensure the interoperability among different value chains, easing the integration and replicability of REDOL overall solutions in different scenarios. As a result, REDOL will reduce the total amount of SUW in Aragon region by re-using, valorising and transforming wastes, by-products and side-streams into new/secondary resources of raw materials of high value for local industries and end-users.

Specific:	Linked Deliverables & KER: D2.4, D5.3, D6.1, D6.3, D6.4, KER13-18
Measurable:	Linked Impact Factor (IF): IF3, IF4, IF8, IF9 (see section 2.1.3)
Achievable:	Linked Partners: LTU, TAT, CAS, MOS, BRI, SUIT, RINA, ACT, ZAR, ALIA, HUSA
Relevant:	Relation to the topic (scope's bullet points): #2" Process (re-)design and adaptation to build a new circular value chain", #3"Investigate the availability and distribution of "waste" resources and

	logistic", #8"Assessment of the economic, circularity and climate benefits"
Time-bound:	Linked work packages and tasks: T2.1, T2.5, T2.6, WP5, WP6, T8.3, T8.6

Specific Objective 4: To establish a **roadmap for the implementation of REDOL solutions** in other urban environments and industrial areas, tackling technical and non-technical barriers.

REDOL will carry out an assessment of the regulatory and social landscape in the Aragon region, as well as current EU standards, to identify barriers and bottlenecks that could hamper SUW valorisation. Based on this landscape and the results from the project demonstration, REDOL will generate recommendations for key stakeholders (e.g. policy makers, waste managers and recyclers) and engage a community of followers to remove these barriers and to support the circular design of the targeted value chains. REDOL will identify and analyse the potential replication of the developed solutions considering their cost-benefit analysis, as well as the technical and regulatory obstacles. Besides, REDOL will connect with the European Community of Practice (ECoP) and other relevant EU bodies for knowledge sharing, dissemination of outcomes, challenges and recommendations.

Specific:	Linked Deliverables & KER: D2.1, D5.3, D7.1, D7.3, KER19, KER20, KER22
Measurable:	Linked Impact Factor (IF): IF6, IF7, IF8, IF9 (see section 2.1.3)
Achievable:	Linked Partners: CIRCE, ITA, UZ, AMS, CEMCH, ICL, EEIP, ZAR, PRA, BOFA
Relevant:	Relation to the topic (scope's bullet points): #6"Define assessment methodologies and evaluate KPIs to measure the performance of symbiosis", #7"Life cycle assessment and life cycle cost analysis", #11"Connect to the ECoP for knowledge sharing"
Time-bound:	Linked work packages and tasks: T2.4, T2.6, T7.1, T7.2, T7.4, T7.5, T7.6

Specific Objective 5: To **assure a successful dissemination and exploitation of the project**, through a strategic and **I-US oriented** commercialization plan, dedicated **business models and key stakeholders' engagement**.

An exhaustive analysis of the market, potential application, risk and opportunities will be performed. Different business model will be produced during the project, integrating circular approaches and agreements for the establishment of symbiotic relationships. This approach will enable REDOL to set up a global business plan that optimises SUW management among the whole value chain key partners, establishing best practices for each waste material, according to technical (technology performance, logistics, etc.) and non-technical (legislative, regulatory, standardisation) aspects. Additionally, REDOL consortium will oversee the involvement of citizens, public bodies and other relevant stakeholders within the project, whereas it will develop a proper communication and dissemination (C&D) strategy of the major innovation outcomes to support the implementation of I-US. To this end, a social innovation spin-off action based on a design thinking approach will be implemented in the region, thus supporting users in the transition from passive consumers towards active citizens impacting the value chains.

Specific:	Linked Deliverables & KER: D7.2, D8.1, D8.2, D8.3, KER21, KER23
Measurable:	Linked Impact Factor (IF): IF4, IF6, IF7 (see section 2.1.3)
Achievable:	Linked Partners: CIRCE, STRA, GEO, ICL, EEIP
Relevant:	Relation to the topic (scope's bullet points): #9"Study social aspects of the community and its improvement through I-US", #10"Create societal awareness (…) highlighting and communicating political and regulatory obstacle between regions/countries"
Time-bound:	Linked work packages and tasks: T2.2, T7.3, WP8

1.1.2. Ambition

The EC highlights the key role of cities in advancing on sustainability and circular economy,[1] as cities must move away from linear systems of resource use and learn to operate as closed-loop circular systems. In order to achieve this objective, REDOL aims at solving the complex challenges behind circular value chains and I-US, not only at **technological level**, but also at **managerial, ecological, regulatory, economic and social** levels. Only by following this approach can hubs for circularity reach a mature enough readiness level, advancing in the SRL (Symbiosis Readiness Level) of the symbiotic interactions among key players. To this end, different innovations will be implemented for each typology of residue and for REDOL as a whole, tackling barriers that affect all the key actors of the value chains. **Digitalisation will be key** to accomplish this objective, as well as the development of new schemes and procedures to **interact between industrial players and with local citizens**.

Starting from the technological perspective, REDOL will deploy innovations along three axis: 1) collection, sorting and classification technologies, 2) recycling and valorisation routes, and 3) formulation of new circular products that enable closing the loop in REDOL value chains. These innovations will be developed and implemented around **5 lines of valorisation**:

Value chain	Packaging (MMPP and P&C)

[1] "Circular Economy Action Plan for a cleaner and more competitive Europe", European Union, 2020

Gaps & Barriers in current SoA	Packaging is an extensive family of entirely different materials. **Multi-material multilayer plastic packaging (MMPP)** enables the industry to minimize the average thickness of packaging while combining functional properties, such as barriers, mechanical strength and heat tolerance. However, MMPP embedded **heterogeneity became a hindering factor** for the recycling industry. Waste characterisation and sorting is key for increasing the recycling efficiency, but conventional MMPP waste management systems are not adapted to identify, separate and recycle MMPP, becoming a key circular economy challenge due to the high volume of MMPP among SUW.[2] Moreover, many packaging is composed of or includes paper and cardboard (P&C), which is **the most recycled material** in Europe, with c.a. 80% recycling rate. Whereas the valorisation technologies are well advanced, developments in **sorting technologies** are needed to **improve the quality of recycled** materials and maximise the value of each fraction to get closer to zero waste.
REDOL R&I	REDOL will improve the classification and separation system for MMPP and P&C wastes managed by HUSA and SAI by means of innovative artificial intelligent (AI) and robotics solutions (CIRCE, AMB) and with the capacity to **identify and separate different MMPP** (IRIS). Specific recycling techniques for MMPP based on **green solvent selective dissolution** will be developed (ITE) and demonstrated (ACT) building on results from **MERLIN** EU project,[3] while recycled polymers will be reused by ACC and TEC to produce circular products like profiles for civil infrastructures, consumer goods, automotive parts and new packaging.
KER	KER2, KER5; KER6; KER12 (see section 2.1)
TRL	Prototypes already available at lab scale (IRIS, ITE) while experiences on sorting and classification by means of robotic systems already exist, but automatic and mobile systems of this nature are still at lab level (TRL5). The development and testing of these novel solutions at HUSA, ACT and SAI will enable REDOL to reach TRL7 and ease its industrialization.
Value chain	**Plastics**
Gaps & Barriers in current SoA	Whatever the application, at the end of their service life, plastic materials are necessary resources that must be recycled. It is estimated that **only 9% is recycled**, and most of it is landfilled or in our environment (land, oceans, rivers, ice and air).[4] The challenge is to increase the amount of plastic waste recycled, addressing the most complex fractions and unsorted mix of plastics, but the **lack of recycling technologies that could upgrade the value of recycled matter is one of the main technological barriers**. Mechanical recycling is the most widespread and mature, although it presents a series of important limitations, while **chemical recycling is a complementary technology, that can help diverting certain plastic waste** which cannot be sustainably (technical or economical) recycled by mechanical processes.
REDOL R&I	Aragon region is already advancing in plastics recycling, including as an example **PLASTICE** HE project which deals with unsorted plastic waste and with whom REDOL will exploit synergies.[5] On this basis, REDOL main challenge is to advance in plastics classification to increase the effectiveness of different valorisation routes, and thus will enhance the traditional sorting and recognition of plastic materials using FTIR spectrometers combined with AI techniques by **deploying integrated smart bins for urban plastics** collection at city level (ICCS and ZAR). Recycling techniques will be developed and demonstrated (MOS) using **green solvolysis assisted with microwave (MW) and ultrasounds (US)** (AIT and CIRCE). Additionally, REDOL will generate **eco-design practices** to facilitate their reuse, remanufacture and recycling (AIT, TEC).
KER	KER1; KER7; KER12 (see section 2.1)
TRL	Consortium partners have tested the sorting and valorisation technologies in previous R&D projects, and prototypes are already available (TRL5). Likewise, the chemical recycling of plastic-based wastes using standard solvolysis technologies have been proved in previous R&D projects (e.g. POLYNSPIRE) and prototypes (TRL 5). Within REDOL they will be adapted to the new conditions of the waste flows and demonstrated at real urban environment (ZAR) and the premises of MOS (TRL7).

[2] Containers/packaging make up a major portion of SUW, 82.2 million tons in 2018 (+28% percent of total generation).
[3] MERLIN: Increasing the quality and rate of MultilayER packaging recycLINg waste
[4] Geyer, R., et al. (2017). Production, use, and fate of all plastics ever made. *Science advances*, *3*(7), e1700782.
[5] PLASTICE: New technologies to integrate PLASTIC waste in the Circular Economy

Value chain	Construction and Demolition Wastes (CDW)
Gaps & Barriers in current SoA	Proper handling and recycling of CDW is a major challenge for the industry due to the **increasing volume of waste** produced (≈375 million tons/year) and its high environmental impact **(CO_2 emissions)**. CDW is a highly **heterogeneous mixture of materials** that typically include metals, concrete, asphalt, mortars, brick, gypsum, wood, cardboard, steel, iron and plastic. As the construction sector is still **one of the least digitalised sectors**, CDW valorisation is far from the wide adoption of automated solutions that could contribute to the creation of novel circular value chains. Enabling management of CDW with advanced sorting systems can lead to less waste generation and management costs at all stages of the construction value chain, contributing to the transition towards a digital and circular economy.
REDOL R&I	REDOL will develop novel automated tools strengthened with robotics, separation and sorting, and material characterisation for better CDW valorisation. A **sensor-based sorter** will be developed (ICCS) for **upgrading CDW waste management** facilities (CAS). C**omposites materials** (≈11% of CDW) will be recycled within the **green solvolysis system** developed under the plastics value chain. Other classified materials will be **assessed within REDOL as alternative raw materials for application in the cement/concrete industry** (VDZ and CEM), formulating new components and products (such as Megalito®), reducing significant CO_2 footprint and increasing resource efficiency. Recovered fibre glass (FG) from composites will be reused to produce **reinforced wood particle boards** (FIN) and **civil engineering infrastructures** (ACC).
KER	KER3; KER8; KER12 (see section 2.1)
TRL	REDOL will start from existing results including AI-based algorithms and know-how available on potential alternative materials to be valorised (TRL5). Further analyses and testing will be required previous to the demonstration in CAS, CEM, FIN and ACC at TRL7.
Value chain	**Textiles**
Gaps & Barriers in current SoA.	The EC strategy on Sustainable Products has flagged textiles as a sector with a **high potential for circularity**. **Automated sorting has great potential for improvement** to sort more efficiently and accurately most textile waste. Moreover, as an alternative to the more widespread mechanical recycling technologies, **fibre-to-fibre recycling processes relying on chemical technologies** that reduce textiles to their basic components need to be further developed to operate at industrial scale.[6] This approach enables to re-spun waste into new fibres, yarns and textiles and increase the sustainability of the sector.
REDOL R&I	REDOL will develop (NTT) and demonstrate (ALIA) a novel **textile waste sorting based on hyperspectral imaging** selection by colour, composition and fabric/garment structure. The sorted textile wastes will undergo novel recycling routes to remove **elastane/lycra** (NTT) which is one of the key challenges for textile recyclers.[7] For mixed plastic-textile fibres, a depolimerization-polimerization process through **glycolysis with ionic liquids** will be optimised (AITEX), followed by **post-condensation and spinning** steps at pre-industrial demonstration level (BRI). Finally, the recycled **polyester yarn will be remanufactured** to produce geotextiles for soil reinforcement (BRI) and surface veil for pultrusion of profile for civil infrastructures applications (ACC).
KER	KER4; KER9; KER10; KER12 (see section 2.1)
TRL	The technologies to be developed currently exist at lab scale (tests performed at max speed 3 m/s for the sorting in NTT, pilot reactor of 100 L and 20 kg/day capacity in AITEX). The advance from TRL5 to TRL7 will be possible by a demonstration campaign in PRA/ALIA and BRI facilities.
Value chain	**Waste from Electrical and Electronic Equipment (WEEE)**
Gaps & Barriers in current SoA	WEEE is now one of the **fastest growing** waste streams including a large range of devices such as computers, fridges and mobile phones at the end of their life.[8] This type of waste contains a **complex mixture of materials**, some of which are **hazardous** (Cd, Hg, Pb, As, oils, etc.) and may cause major environmental and health problems. In addition, many WEEE contain **rare and expensive resources** (Au, Ag, Pd, Pt, etc.), which can be recycled and re-used if the waste is well managed. However, **barriers on transport and processing costs** still prevent their full valorisation.

6 Köhler, A., et al. (2021). *Circular economy perspectives in the EU Textile sector*. Publications Office of the European Union.
7 P. Harmsen, et al (2021). Textiles for circular fashion: The logic behind recycling options. *Sustainability*, *13*(17), 9714.
8 O. S. Shittu et al., Waste Management, 120, 549-563.

REDOL R&I	REDOL will work both in the **automation of the Print Circuit Boards (PCB) separation process** (CIRCE) and on the development of novel processes for the **recovery of precious metals and rare earths from PCBs** (RINA) by replacing conventional acids with the formulation of novel deep eutectic solvents (DES). In addition to the demonstration of these technologies at TAT, REDOL will make use of the **casing plastics protecting WEEE** and discarded in the valorisation process to substitute **flame-retardant fillers** like aluminium trihydrate in pultrusion process (ACC).
KER	KER5; KER11; KER12 (see section 2.1)
TRL	RINA has experience in metal recovery by DES with different wastes materials and with DES formulations (TRL5). The application of this know-how in the development of a novel process for PCB valorisation to be demonstrated in TAT will allow REDOL to reach TRL7.

In order to successfully implement these innovations at the SUW circular value chains, REDOL will put in place a set of **horizontal activities** that will be facilitated through the **I-US REDOL's Platform**. This platform will put in place a multi-criteria evaluation framework for industrial symbiosis, including data sharing mechanisms and enabling the role of facilitators in the setting up of new initiatives. The adaptation to REDOL will enable the analysis of waste streams and boundary conditions to deploy the I-US concept, as well as competences, stakeholders and regulations required for its successful application. The platform will be designed to provide new circular business models and analyse their environmental, economic and social impacts from a life cycle approach.

Additionally, the **logistics of the waste streams** and inverse logistics of the valorised products will be analysed and optimised. The visualization of the containers, trucks and waste flows will be made available by means of **advance sensing technologies**, installations of **novel smart urban bins** and **GIS applications**, which will result in a **digital twin of Zaragoza's waste management** system. These innovations will be complemented by **social initiatives** promoted by Zaragoza's City Hall and main RTO and waste management institutions, with the aim of involving and improving the behaviour of citizenship to adopt sustainable practices linked to waste management. The overall ambition and R&I maturity of these horizontal developments are summarised in **Table 1**.

Table 1. TRL development along REDOL project.

Development	Current TRL	Final TRL
I-US Platform	**TRL5**. Platforms on IS and waste management exist but must be adapted to the needs of I-US and the region particular needs.	**TRL7**. Ad-hoc tools and models integrated; I-US Platform tested in REDOL value chains using data from all key actors.
Inline sensing for traceability	**TRL5**. Smart inline sensors validated for different sectors (e.g. DISIRE project). There is a need to adapt them to REDOL SUW streams.	**TRL7**. Field evaluation at the addressed SUW value chains in Zaragoza, data exchanged in real time and a secure way with the platform.
GIS based mapping	**TRL5**. Even though GIS is a mature technology, it must be implemented in the city's waste management procedures.	**TRL7**. GIS tool for mapping resources and ensure SUW traceability integrated in the platform and with REDOL sensing system.
Logistics optimization	**TRL5**. Optimization of delivery routes is common for the transport sector but must be incorporated to SUW (inversed) logistics.	**TRL7**. Optimization algorithms validated by using REDOL value chains and key actors information through the I-US platform.
Forecasting and s-LCA tools	**TRL5**. Models exist in current SoA, but there is currently a high uncertainty in the availability and price of waste resources.	**TRL7**. Currently available models and algorithms will be trained with real data and included in the platform and I-US agreements.
Circular eco-designs	**TRL5**. Solutions for designing new products with a better end of life formulated.	**TRL7**. Implementation of the solutions formulated in real circular products.
I-US guidelines	**TRL5**. In EU, initiatives on IS and hubs for circularity are currently in the process of being implemented or providing first experiences.	**TRL7**. REDOL will generate a handbook and recommendations for the implementation of hubs for circularity based on I-US real experiences.
I-US business models	**TRL5**. Novel business models on CE and IS are in the process of being tested in EU projects such as CIR4LIFE and PLASTICE.	**TRL7**. Existing and new business models applied to the context of I-US and agreed among key actors of REDOL value chains.

1.2. Methodology

1.2.1. Overall concept and approach: Aragon as a Hub for Circularity

REDOL project (meaning *Circular project* in Aragon regional dialect) has been conceived to **advance the SRL of Aragon's Hub for circularity** in order to demonstrate the set-up of new business models relying on SUW as alternative resource for local industries. Aragon is a European NUTS 21 region that has set an ambitious initiative to **reach an innovative circular economy by 2030, enabling the full valorisation of residues**. Main activities are being deployed in the capital, Zaragoza, which aims to become one of the first 100 climate neutral cities by 2030 and has already put in place projects such as Circular Biocarbon, the first urban biorefinery aiming to **valorise 100% of the**

city's organic residues. REDOL will therefore complement these efforts by advancing in complementary SUW streams addressing **5 out of 7 of the key value chains defined by the EC Circular Economy Action Plan** (electronics and ICT, packaging, plastics, textiles, construction and buildings).[9]

Previous to the development of this proposal, **the implementation of a municipal-industrial symbiosis approach in Zaragoza was assessed based on scientific and commercial data**.[10] This assessment provided a set of strengths and weaknesses to be exploited/overcome within REDOL project. Main strengths include: i) Aragon generates a **rich mix of industrial, urban and rural wastes** with the ideal volume and seasonality to implement effective CE; ii) The region counts with public and private entities leading innovative initiatives for recycling in the **common framework of Aragon Circular initiative**; iii) Zaragoza meets the European recycling target and **counts with dedicated facilities for SUW treatment** in contrast to similar municipalities; iv) **Initiatives to incentivise sorting and recycling in households and business are already on-going**. Key weaknesses to be addressed include: i) There are **data limitations and lack of traceability**, which makes sensing and digitalisation crucial within REDOL; ii) Almost 70% of SUW ends up in the mixed fraction due to the **saturation and inefficiency of current collection and sorting systems**, which implies the need for REDOL to implement innovations in this step; iii) Policymakers together with manufacturers and recyclers still need to **put in place governance structures** that allow, e.g. to define strategies for promoting the use of low impact materials and standards on recycled products that facilitate reuse and provide information to improve recycling processes.

Taking into account the results of this analysis, in REDOL, the circular management of SUW will be done through coordinated actions divided along four pillars: **1) to upgrade waste manager´s facilities with new technologies** (i.e. sorting, processing/recycling techniques), **2) to design new circular products** (valorisation of secondary raw materials and implementation of eco-design practices)**, 3) to improve management and data sharing procedures by implementing digital tools** (e.g. smart sensors, IoT, optimization algorithms, central I-US Platform), **4) to generate a vibrant ecosystem around circular and symbiotic value chains** (i.e. to improve the organizational skills of local key actors and to engage and unlock the potential of local communities towards innovative circular schemes). The resulting value chains obtained from REDOL successful execution are depicted in **Figure 2**.

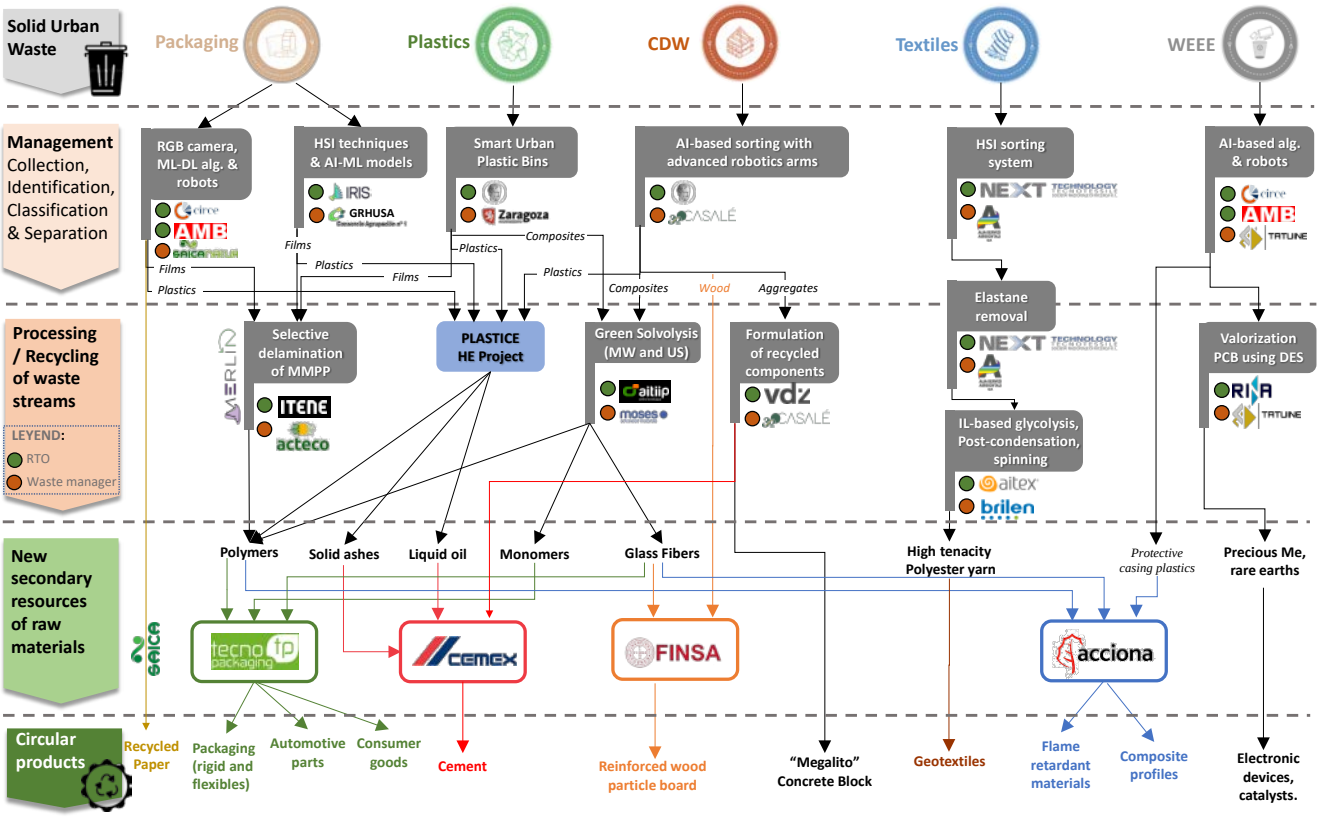

Figure 2. Overall scheme of REDOL concept and approach.

Prevalent computer vision algorithms for SUW sorting will be examined from aspects like waste sources, task objectives, application domains and dataset accessibility. REDOL will implement new shifting trends by using machine learning and deep learning algorithms. The robustness of the developed digital tools will be enhanced by advanced computation powers and tailored algorithms for waste value chains such as plastics, packaging, CDW and textiles. REDOL preliminary studies will be based on environments and artificially collected data. Further research

[9] "Circular Economy Action Plan for a cleaner and more competitive Europe", European Union, 2020

[10] Valero, A., Torrubia, J., Anía, M. Á., & Torres, A. (2021). Assessing Urban Metabolism through MSW Carbon Footprint and Conceptualizing Municipal-Industrial Symbiosis—The Case of Zaragoza City, Spain. *Sustainability*, *13*(22), 12724.

efforts will consider the complexities of real-world scenarios and implement digital techniques in industrial waste sorting practice. REDOL will also call for open sharing of waste image datasets for interested researchers to train and evaluate their AI-based algorithms.

In the following pages, REDOL implementation pillars will be described in detail, addressing facilities upgrade of waste managers and industrial partners (Section 1.2.2), generation of secondary raw materials (Section 1.2.3), digital approaches for improving SUW management (Section 1.2.4) and managerial/social actions to be implemented at local level (Section 1.2.5). Accordingly, REDOL will not only provide an environmental and technical solution to SUW management and valorisation, but also a managerial and economic one, **generating viable models that will boost the competitiveness of the different industries involved**. Such an approach has been designed to transform Zaragoza City and Aragon Region into a **Hub of Circularity of reference for EU cities** willing to follow CE practices. Accordingly, the project will provide a **handbook for the establishment of hubs for circularity**, aligning the project activities with initiatives such as the ECoP and will start a **first replication stage in 3 EU cities** (Section 1.2.6) with whom REDOL will exchange best practices and results.

1.2.2. Waste managers and REDOL use cases

HUSA Waste manager: Improvement the sorting system of a packaging classification plant

►**Waste manager description.** The HUSA packaging selection plant separates different types of light packaging from collection carried out by citizens. The plant is a 2,280 m² warehouse, with a treatment capacity of 2 t/h. In 2021, 6,200 tons of packaging were processed. It automatically separates the unloaded containers into 7 fractions: steel, aluminium, PET, PEAD, film, mixed plastic and multilayer beverage carton.

►**Technology SoA**. The commercially available solutions (near-infrared spectroscopy, NIR) generally focus on identifying mono-materials of a few selected types (PE, PP, PET, etc.). However, MMPP identification is limited because of the thickness of the packaging and the composition of the different layers. Therefore, many recyclers are still sorting plastics by hand, which is unreliable and expensive, or this multilayer plastic cannot be identified, and consequently recovered.

►**REDOL Innovation.** REDOL will go beyond the state of the art by adopting **hyperspectral imaging** (HSI) together with **AI-Machine learning (ML) models**, enable bridging practical challenges previously not feasible in terms of **classifying MMPP**. An imaging technique such as HSI is preferable for MMPP identification of small films or shredded material fed on a belt. Additionally, AI statistical evaluation, implemented using a custom-built state-of-the-art platform, will allow to reach sufficient accuracy and speed in the material detection to be implemented in the real environment.

►**Main goals to be achieved through REDOL.** IRIS will develop a real-time monitoring system for MMPP identification, that will be validated at HUSA, improving the effectiveness and the quality of the output material.

Use Case #1: Delamination of MMPP structure by selective decomposition of polymer layers (ACT)

►**Waste manager description.** ACT is a SME dedicated to the management and recovery of waste, recycling hangers, fridges, yogurt cups, bottle cups or even car bumpers (380.000 t/y). They recycle +12,000 tons of packaging/y, for which ACT target finding a solution to valorise MMPP.

►**Technology SoA.** For delamination of MMPP structure there are processes focused on the selective dissolution and precipitation of one of the polymer layers. This is solved selectively, removing later the non-soluble components and precipitating the polymer using a non-solvent media. These routes achieve the delamination by depolymerization of one of the layer, PET or polyamides (PA), e.g, using sub and supercritical H_2O.

►**REDOL Innovation**. Green solvent-based processes will be specifically designed to **dissolve the required layer in MMPP waste**, with different approaches. Green solvents, such as **CO_2 in supercritical conditions** will be used to selectively dissolve (i) the adhesives between layers or (ii) specific layers such as the tie layer. In addition, it will study the influence of the presence of other internal layers in the structure such as primers and/or functional barriers (e.g., etilen-vinil-alcohol) in the process.

►**Obtained secondary raw materials.** Different **polyester (PES) and polyethylene (PE)** fractions in the structures will be tentatively separated and recovered. The produced secondary polymers could be used in the plastic injection industry, e.g., hangers' injection, automotive parts, yogurt cups.

Use Case #2: Overcoming barriers in sorting and classification (SAI)

►**Waste manager description.** The Saica Group is one of the leading companies in EU in the manufacture of recycled paper, with an annual production of 3.3 million tons. Saica Natur Zaragoza process in its facility a total of 125.000 t/y. The developments carried out in optical classification technology allow SAI to improve the quality of the materials that they provide, maximizing the value of fractions and contributing to "zero waste".

►**Technology SoA.** SAI currently has a semiautomatic sorting line and a manual picking station. Both systems have the purpose of removing non-paper components from the wastepaper received from local businesses and household.

By doing this sorting, the paper mill generates less residue in the pulper during the paper making process. This pulper residue is a very difficult residue to recycle worldwide, and its destination is normally landfill or energy valorisation.
►**REDOL innovation.** A novel **automated mobile sorting line** for paper and packaging classification will be developed and demonstrated at SAI. The system will include optical equipment based on **HSI and RGB cameras** and real-time sensors for non-paper components. They will use **ML and deep learning (DL) algorithms** applied to the different spectrums images (CIRCE) and giving instructions to the pickling robotic elements (AMB). The line will also include ancillary's services (e.g., feeder system, separation box, etc.) and **flexible robots that will allow it to be deployed at other SUW lines** addressed in the project (like WEEE at TAT).
►**Main goals to be achieved through REDOL.** The developed sorting system will allow to improve the quality of the recycled paper produced in SAI. It is estimated that we will apply the REDOL technical solution in 10% of total managed volume, that is the one with the highest levels of non-paper components.

ZAR Waste manager: Deployment of smart urban bins for plastic collection at citizen's level

►**Waste manager description.** ZAR manages post-consumer plastics coming from household wastes, which are collected as part of mixed municipal SUW at the Complex for the Treatment of SUW in Zaragoza (CTRUZ facility with a processing capacity of mixed SUW of ~ 250,000 t/y). The actual recovery of LDPE (colour and white film) at CTRUZ is about 2,500 t/y, while potential PP and PS waste streams are about 1,400 t/y and 460 t/y, respectively.
►**Technology SoA**. Multiple manual and automatic sorting systems utilizing different identification methods have been developed.[11] Among them, NIR technology is primarily used, whereas alternatives like FTIR spectroscopy allow to obtain more precise chemical information. To facilitate the sorting of urban mixed plastics, advanced learning techniques should be developed to empower categorization tasks and identify hidden patterns in spectrums, increasing the classification accuracy and reducing the inference time. This will facilitate to provide real-time recognition response capabilities to function on mechanical urban sorters with robotic-based mechanism.
►**REDOL Innovation.** A multi-spectrum analysis will be performed allowing a **smart waster sorting** mechanism to accurate separate **urban mixed of plastic in real-time**. Micro-electro-mechanical-system FTIR spectrometers will be used to scan the disposed objects aiming to classify them in **polyethylene terephthalate (PET), Polyphenylene Ether (PPE), Polyethylene (PE)** and others. ML techniques based on convolutional neural networks will be developed to reduce the inference time and ensure the proper classification. Steppers are used to move the objects towards the buckets and then an air-basedPA system is used to push the object in the appropriate bin. **Lidar sensors** will be used to measure the level of the bins aiming to **optimize logistics operations**.
►**Main goals to be achieved through REDOL.** The proposed mechanisms will strengthen the valorisation of the waste by transforming the existing streams to improved -in terms of quality, purity, safety, usability and patterns identified- and new ones -by creating new flows of secondary materials driven by the technology innovation based on separation that was not possible before- and by facilitating their improved transportation and the context in which citizens recycle plastics.

Use Case #3: Green solvolysis assisted with MW and US for recycling composite materials (MOS)

►**Waste manager description.** MOS is equipped with most of the plastic transformation technologies (injection, rotational moulding, blow moulding, thermoforming, injection moulding, etc.) devoted to creating material that meets the most demanding requirements for plastic products. MOS offer advanced materials and products for packaging, urban waste collection and automotive sectors, combining recycled and bio-based materials. Through REDOL, MOS ambition to create an effective after-use plastic economy, in particular for high technical plastics: thermoset composites and for thermoplastic high-tech based composites. High costs of these technical materials are making chemical recycling of the monomers a huge opportunity for MOS business development.
►**Technology SoA.** The composite waste treatment available techniques that are already commercial or are approaching a commercial status can be divided into mechanical, thermal, chemical and biological.[12] In a chemical recycling process (solvolysis), the matrix is disintegrated into a chemical solution. Compared to thermal technologies, solvolysis requires lower T to dissolve the resins, resulting in a lower damaging of fibres. However, it incorporates salts or other modifying components that damage fibres and can be highly polluting (conventional solvents account for about 20% of the volatile organic compound emissions).
►**REDOL Innovation.** 1) Improve the **separation of the components** (**resin and fibre**) through energy efficient technologies, assisting with **US** (increasing 40% the current separation rate and decreasing in a 26% the operating time) **and MW** (decreasing 30% the energy demand).[13] 2) **Test and adapting novel green solvents** (e.g., biobased polar protic and aprotic) to replace the usually toxic and flammable organic solvents, reduce chemical waste and improve the safety. AIT and CIRCE will take advantage regarding their knowledge in other recycling technologies

11 S.P. Gundupalli, et al, Waste management, 60, 56-74.
12 W. Post et al. Polym. Rev. 2020, 60, 359.
13 PolynSPIRE H2020-NMBP-SPIRE-820665; develop. of a solvolysis process MW-assisted for polyamide and polyurethane

(mechanical and thermal) to use them as pre-treatments for the solvolysis process to study efficiency in combinations and potential synergies.
►**Obtained secondary raw materials.** The innovative recycling technology will provide high quality of recycled fibres (mainly **glass fibres**) which will be directly integrated in current MOS products (lids for urban waste skips and bumpers for automotive sector) and **chemical building blocks** that will be the base of recycled polymers.

CAS Waste manager: Deployment of AI-based sorting system with advanced robotics for CDW

►**Waste manager description:** CAS is the owner of a CDW waste management plant, located in the Zaragoza with a surface area of 60,000 m^2. In 2021, CAS managed c.a. **80,000 tons of waste**, 50% as unsorted wastes. The classified waste that they cannot take advantage of, such as plastics, wood or cardboard, is taken to other authorized waste managers, whereas the obtained recycled aggregates are incorporated back into the construction sector in the form of "Megalito", a precast concrete block made with 100% recycled aggregates.
►**Technology SoA**. Different automated CDW sorting systems have been developed. Among them, HSI technology with successive projections algorithm, extreme learning machine, particle swarm optimization and online learning, achieves around 90% detection accuracy.[14] Although the existing sorting algorithms offer accurate results, new ML/DL techniques must be utilized to sort a wider range of CDW, offer higher reliability and enable real-time recognition responses capable of operating on robotic-based mechanical sorters.
►**REDOL Innovation.** A multi-spectrum analysis will be performed allowing a **robotic-based sorting mechanism to accurate separate CDW in real-time** (ICCS and CIRCE). The classification will be relied on ML/DL approaches and the separation will be based on robotic arms, upgrading the existing processes. AI will also facilitate to identify hidden patterns in the component analysis methodologies. To easily classify the CDW without processing limitation, the industrial camera will capture the Region of Interest, whereas the hyperspectral camera captures the spectral information. Using **3D vision systems**, the position and the shape of the wastes is identified within the conveyor to guide the robotic arms for picking and placing the waste in the right bin. Based on the extracted spectral feature bands, ML will classify the CDW in one of the following categories: **glass, ceramic, concrete, wood, gypsum, metals, other or mixed materials**. Additionally, the identification of **plastics** (4-55%) **like PVC, PET, PE** will be performed by multispectral cameras which will analyse the emissivity of the different polymeric components. Finally, the robot will pick the object and place the object to the individual container.
►**Main goals to be achieved through REDOL.** The proposed mechanisms will strengthen the valorisation of CDW by transforming the existing streams into improved (quality, purity, safety and patterns) and new ones, creating new flows of secondary raw materials that may be used in upcycle solutions by other industries.

ALIA Waste manager: Novel textile waste sorting system and elastane removal

►**Waste manager description:** ALIA is a management company of environmental services in Tuscany (IT). Its goal is to promote good waste management practices, promoting separate collection and safety in the management of plants. It includes 6 plants in the province of Florence (49 kt/a), Prato (130 kt/a) and Pistoia (26 kt/a).
►**Technology SoA**. New technological solutions are needed for textile waste sorting in search of higher flexibility. Among the most promising technologies are NIR, HSI and radio frequency identification (RFID). However, items with a lining or multilayer fabric cannot be sorted correctly with these techniques and algorithms are only suitable for the most commercial blends. Also, the presence of elastane (lycra) prevents the optimal downstream recycling.
►**REDOL innovation.** A **novel textile waste sorting based on HSI selection by colour, composition and fabric/garment structure** will be developed (NTT) and demonstrated (ALIA) with real mixed fibres blend flows. NTT will also validate an innovative technique to **separate elastane (lycra) from textile fibres** with a closed cycle solvent leaching for the next chemical recycling.
►**Main goals to be achieved through REDOL:** 1) Improve the textile waste sorting efficiency to provide the required inputs/quality to downstream recycling technologies, and 2) Remove 100% of elastane fibres from natural and synthetic blends for the following chemical/mechanical recycling processes.

Use case #4: Chemical recycling of textile wastes by IL glycolysis followed by spinning (BRILEN)

►**Waste manager description:** BRI is a high tenacity polyester (HT PES) yarn manufacturer (extrusion, spinning, drawing and winding). When recycled HT PES yarn is produced, current inputs are post-consumer PET from bottles mechanically recycled. Waste is not homogeneous, and properties of the final product are not as stable as could be with an original PET chip. PES substandard yarn can be recycled with the textile waste already managed by other companies of the consortium.
►**Technology SoA**. Mechanical recycling approaches is the most employed at present in the textile sector. However, textiles cannot be endlessly recycled mechanically without reducing their properties and quality. Chemical recycling methods enable 100% breakdown to monomer, then the textile could be recycled an infinite number of times.

14 W. Xiao, et al., *Plos one*, *14*(1), e0208706.

However, results of LCA show that not only chemical recycling incurs greater costs, but it also contributes to an overall negative impact in terms of a larger net consumption of energy and environmental effects.
►**REDOL innovation.** This will be the first time a chemically recycled polymers derived from textiles by **IL-based glycolysis** is used to produce HT PES yarn in Spain. **Post-condensation** is focused on increasing the viscosity of the polymer supplied by AITEX to comply the characteristics **to be spined as a HT PES**. Product has less than 2% of other substances, which makes the yarn almost 100% recyclable.
►**Secondary raw materials:** HT PES yarn is a technical product used as reinforcement, giving strength to structures. A **geotextile made of HT PES** with high mechanical resistance and drainage capacity will be produced.

TAT Waste manager: Deep eutectic solvent replacing acids for recovering precious metals from WEEE

►**Waste manager description:** TAT is a SME located in Aragon specialised in the management and treatment of WEEEs. They recover strategic metals from PCBs derived from urban residues like computers, servers, digital devices, phones, etc. Other waste flows derived from WEEE that they manage are glass foundry wastes (3 t/y), PVC plastics (24 t/y), acids (4 t/y) and plastic protective cases.
►**Technology SoA**. The current process for recovering PCB metals is based on thermochemical and electrochemical treatments, using acids (e.g., HNO_3, H_2SO_4). After concentration, the strategic metals in the form of sludge are chemically treated in TAT laboratory. Conventional solvents exhibit aggressive profiles that may lead to health, environment and safety related issues, as well as damage to industrial equipment.
►**REDOL innovation: Deep eutectic solvents (DESs)** have emerged and attracted increasing. DES refers to a mixture of two (or more) compounds, a hydrogen bond donor, and a hydrogen bond acceptor, which has a lower melting point than the individual compounds. DES share common properties with ILs, but they are more cost competitive, and synthesis and purification steps are easier. **DES-based technology for PCB metal recovery** will be developed by RINA for different species and under different DES formulations. The activities will be carried out at laboratory scale (5-10 kg per batch) up to the layout of treatment line at TAT plant (supported by CIRCE).
►**Secondary raw materials:** PCB recycling will allow to recover species like **Cu, Fe, Ni, Au, Pd,** which can be reused for manufacturing new electronic devices o in the chemical industry (catalysts). Additionally toxic and hazardous metals like Hg, Pb, Cd, Cr will be prevented from posing an environmental hazard.

1.2.3.End users valorising obtained secondary raw materials

REDOL will provide solutions for designing **new circular products** with better End of Life options based on the combination of results and generated secondary raw materials. Through the project execution, secondary raw materials will undergo a final analysis to understand how they can be reintroduced in the market, while a specific task (T8.3) has been included to engage end-users in the acquisition of these materials, closing the circular value chains. For secondary raw materials that require further treatment or modifications in the production processes, a set of End-users are involved in REDOL to demonstrate the feasibility of producing new circular products:

End-user #1: Fabrication of fibre glass reinforced wood particle boards (FIN)

►**End user description:** FIN is a Spanish company focused on the manufacturing of wood particle boards. In their plant in Aragon, FIN manages 36.000 t/y coming from domestic or industrial furniture. The use of recycled wood residue in current particleboard production is limited, otherwise the particle boards do not have the required quality. Modification with nanocellulose or the use of bio-based products has also been proposed, as well as hidrofugant additives, minimizing the loss of physical and mechanical properties and improving stability.
►**REDOL innovation:** FIN and ITA will incorporate into the mix of wood particles used to **manufacture the particle boards GF** coming from composite wastes. The main objectives pursued with this particle addition are: (i) to **increase the mechanical properties**; strength, stiffness, keeping constant surface strength and screw pull out resistance, (ii) to increase the **water resistance**, by reducing the swelling after and (iii) to increase the **thermal conductivity** and the fire resistance. FIN and ITA will develop a feasibility study of compatible mixing: organic particles (wood chips), inorganic particles (FG chips or yarn) and inorganic or organic matrix (glue and adhesive).
►**Circular products:** The **new glass fibre reinforced wood particle board**, 100% coming from recycled waste, will increase current market share of FIN in terms of tons of wood particle board and will involve the creation of a new product to be incorporated to the current portfolio of the company.

End-user #2: Civil engineering applications using secondary raw materials (ACC)

►**End user description:** ACC is a greenfield developer and operator of energy & infrastructure that provides solutions to SDG. ACC Construction Technology and Innovation division's polymer and composites group are pioneers in development of innovative solutions for challenges like corrosion, lightweight and easy installation. Some of these innovations have won awards as best innovation at JEC world-France in 2010, 2011, 2016 and 2017.
►**REDOL innovation:** In line with its sustainability principles, ACC within REDOL aims to provide 3 user cases:

- Together with AIT, ACC will incorporate fibre recovered from solvolysis process in development of **new composite profiles** such as flats, beams, I sections, Box sections, etc. ACC will manufacture and validate the design profiles using resin infusion process by industrialized process.

- Together with AITEX, ACC will incorporate FR surface veils using recycled fibres (from AIT/BRIL) using pultrusion process. The main objective is to obtain a **surface FR veil for manufacturing composites profiles** meeting stringent FR requirements of construction industry for example railway infrastructure applications.
- ACC will also target to demonstrate that **WEEE plastic cases** derived from TAT in form of powder could be used as secondary raw material to substitute FR fillers like ATH (primary raw materials) in pultrusion process.

►**Circular products:** Through these use-cases, ACC aims one hand, improve the tenders score by increasing social and environmental impact, and in the other hand increase its portfolio of green composite solutions and its market share in geographies which gives preferences to environmentally friendly sustainable solution.

End-user #3: Energetic and material valorisation of reject flows in the cement industry (CEM)

►**End user description:** CEM Spain Operations, S.L.U. is the owner of a clinker and cement production plant located in Aragon. CEM has been promoting for years the use of alternative raw materials in the manufacture of cement to reduce the CO_2 emission factor.[15] They target achieving 25% substitution with decarbonated raw materials in the manufacture processes.

►**REDOL Innovation:** CEM will study the feasibility of energetic and **material valorisation** of certain waste streams derived from the addressed value chains that are currently mixed and/or cannot be used. The applied upstream IA-based separation (e.g., CDW in CAS) will facilitate the characterization and analysis of these fractions as potential component of the **clinker production process** e.g., (aggregates, sludges, ashes from plastic pyrolysis, etc.). On the other hand, CEM will also perform an **energy efficiency study** over other waste/by-products derived from SUW **to substitute fossil fuel in the clinker oven**, e.g., pyrolysis ashes, textiles, paper,

►**Circular products:** 1) Use of SUW derived by-products as raw material to produce clinker in substitution of limestone ($CaCO_3$) and obtained greener cement for the construction industry with much lower carbon footprint. 2) Use of rejected flows from addressed SUW value chains as alternative fuels in the clinker oven.

End-user #4: Packaging and plastic-based circular products (TEC)

►**End user description:** TEC main purpose is conducting R&D on advanced polymeric materials and their transformation processes for packaging and industrial plastic applications. TEC develops all its activities keeping in mind an eco-innovative perspective, analysing in each case the different alternatives that would reduce the environmental impact of both their products and/or their processes.

►**REDOL innovation:** In REDOL, TEC will propose a set of economically viable technologies that will recover materials with high purity and performances like **glass fibres, polymers/monomers from thermoplastics** (i.e., Nylon, ABS, PP, PC, PET and PS) and thermosets after solvolysis. TEC will demonstrate their use in new very stringent value chains and applications such as **automotive, aviation or construction** to ultimately reduce the use of virgin materials. This will help alleviating Europe's external dependency on virgin resources and will open up new business opportunities and revenue flows for recycling companies. Different grades of recycled materials will be defined as direct substitutes of current conventional materials with the aim of enhancing and strengthen plastics secondary market. Accordingly, testing procedures and trials will be run to guarantee the required quality.

►**Circular products:** Plastic products will be produced for flexible (film) and rigid (packaging, consumer goods, automotive, etc.) applications. The final application will depend on the final quality of the recycled material.

1.2.4. Advanced digital technologies for SUW management and valorisation

REDOL will rely in the **application of digital technologies** to (i) increase the quantity and quality of recovered materials and (ii) overcome the barriers for implementing industry 4.0 concepts. Main developments include:

- **AI and ML** will allow to increase the knowledge on the composition of SUW, providing a continuous mass balance of the entire process for the digital models developed within REDOL.
- **RFID systems**, which allow greater traceability of all selected materials, better management of warehouses and stocks and, ultimately, optimization of the logistics.
- **Dual systems of AI and advanced robotics** to improve the capacity of waste selection, obtaining a greater number of materials, of better quality and improving the working conditions and well-being of workers.
- **Data processing, intelligent software and smart in-line sensors**, from suppliers to customers, enabling to forecast, control, plan and produce intelligently, thus generating greater value for the entire value chains.

The higher availability of data will be used in the design and development of a methodology for **monitoring the life cycle impact of the value chains,** as well as a **decision-making system for improving reverse urban freight transportation**. The measurement and improvement of SUW reverse logistics in the hub for circularity will be performed attending to the environmental, economic and productive areas of sustainability. As a result, **good practices will be established to reduce the cost of SUW return while reducing environmental impacts and collection congestion**, improving quality of life.

[15] Proof of this is the project within the framework of the CEMEX Sustainability Chair with the UZ.

Overall advancements in digitalisation will be integrated into REDOL I-US platform, which will enable the **I-US data management for operation in a real industrial environment**. The basic idea behind this tool is to analyse and simulate the operation in I-US systems, allowing the evaluation, simulation and operation of waste resource streams to achieve a progressive CE. By using real-time models of different parties involved in the I-US system, it will be possible to understand how processes will be affected as a result of an extra-variable addition and thus, it **enables the optimization of the raw materials and energy flows**. Therefore, this platform will act as a **virtual intermediary between partners**, reproducing the characteristics of a real system while ensuring a secure data sharing channel. **Key features of the I-US Platform will include**: 1) Matching the requirements of the industries in real time (availability and quality of the resources); 2) Managing logistic procedures (storage and transport), optimizing the capacity of intermediaries and end users; 3) Integrate models and optimization algorithms to ensure the correct development of industrial processes; 4) Forecast and optimize the behaviour of waste streams.

1.2.5. Implementation of new procedures: Change of mindset

The success in the implementation and long-term operation of REDOL value chains will not only rely on the readiness level and effectiveness of the technological innovations but will also depend heavily on the **awareness level of the involved entities and the willingness and trust of companies** to engage in this cooperation. For this reason, REDOL will work in the creation of joint management structures and the establishment of fluid communication channels. Generation of trust among the parties will be also ensured by the development of **data sharing modules that enable a secure exchange of information along the value chain**. REDOL will also step forward in the **definition of the main business models' elements to facilitate the creation of win-win relationships and a fair share of benefits between parties**. This comprises the settlement of price schemes for the products/services exchange, the definition of ownerships models for common infrastructure and services, schemes for risk management plans and compensation clauses between parties to minimize uncertainty of the symbiosis case and the support in attracting private and public funding. Moreover, improving the efficiency of collection and sorting systems requires the **proactive participation of citizens, consumers and waste managers**. In this sense, strategies combining social awareness campaigns, selective collection, smart classification and economic instruments (e.g., reduction waste management taxes) will be approached. For these strategies to be effective, citizens must be involved from the very beginning, so **a design thinking approach will be implemented**. Social innovations within the project will be complemented with on-going initiatives at regional level, which are framed within Aragon Circular initiative for the promotion of entrepreneurial activities. As a result, **REDOL will create a Circular Community in which knowledge sharing activities and co-creation sessions will be promoted**, constituting the seed for the further growth of the value chains while closing the loops.

1.2.6. Urban areas for REDOL first stage replication

● **Bornholm (DK)** is an island with administrative status as a Regional Municipality with a population of ≈ 40,000 inhabitants. BOFA is the waste management authority and service provider and is part of the Regional Municipality of Bornholm. It operates waste management infrastructure e.g., a waste incinerator plant, landfilling facilities, recycling stations, etc. All waste is managed by BOFA, with approximately 75,000 t/y. Around 60-65% of waste is sent for recycling, 30-35% is incinerated and the remainder is landfilled. BOFA has participated in previous EU projects such as the newly initiated LIFE-IP project "Circular Economy Beyond Waste" that will support together with REDOL the transition towards CE in the island.

►**Main goals to be achieved through REDOL**: As part of Bornholm's zero waste vision for 2032, BOFA is keen on any participation in the proposal to inform its strategic choices and set-up of initiatives. In particular within the current transition to a 12-fraction waste collection system for households, it is hoped that the project can support increased recycling of beverage cartons (MMPP) from the organic fraction. Previous Living Lab data has quantified the amounts of these factions from households to be approximately 8 kg per household per year.

● **Prato (IT)** is a city located in Tuscany, with more than 195,000 inhabitants. Historically, Prato's economy has been based on the textile industry with its famous textile district, which represents about 3% of EU textile production. Thanks to its textile recycling techniques, Prato is considered one of the most innovative industrial cities in Italy. Also, Prato was the first industrial district to develop a closed water cycle to reuse wastewater. It has also limited the CDW by recycling and reusing various materials.

►**Main goals to be achieved through REDOL**: The results from REDOL will aim at establishing new SUW value chains at Prato by optimising the exploitation of local resources. REDOL knowledge transfer will enable to map all existing I-US potential activities in the city, engage local actors to guarantee the application and mutual learning process for CE, and develop a stronger governance model with all the actors involved at local, national and EU level. Prato main players and potential beneficiaries will be part of activities of cross-border cooperation and dissemination (WP7 and WP8).

● **Amsterdam (NE)** city includes the creation of CE one of its sustainability pillars. Amsterdam is a frontrunner in sustainability and circular thinking and has already in place targets for 2030 (halve the use of primary resources) and 2050 (become 100% circular). In this sense, AMS is the institution within the Metropolitan Area that addresses the

urban challenge of transitioning from a linear to a circular model of resource management. The whole metropolitan area has a population of about 2 million people and generates 398 kg/capita with a low separation rate (14%), of which 18% is recycled and 82% valorised for energy generation, thus achieving a zero landfilling.

►**Main goals to be achieved through REDOL**: The main objective of AMS in REDOL is to increase the recycling rate by providing higher added value to the recovered and classified waste. To this end, the city has already identified 3 key value chains: Organic, consumer goods and CDW. Accordingly, AMS expects to learn best-practices from ZAR on the implementation of an urban bio-refinery (CIRCULAR BIOCARBON) and the establishment of approaches in CDW, which is expected to have an impact in materials savings of 500 kt, decrease GHG emissions in ½ Mt and generate 700 jobs. Moreover, waste separation and return logistics are currently considered as a weakness, so AMS is interested in replicating REDOL digitalisation tools addressing this issue.

1.2.7.Methodology

REDOL concept relies on the implementation of a CE approach for the whole lifecycle of SUW. Circularity principles require the integration of the entire value chain, the implementation of digitalized sorting and classification of waste and an appropriate management and recycling to be able to valorise the secondary products. Having this in mind, REDOL methodology has been developed in the following phases (P):

P1 – Data gathering, characterization and alignment between the value chain main actors

Main Activities	• To define and collect key information to be exchanged. • To characterise value chains and key players interaction • Establishment of the baseline for a participatory approach with citizens and stakeholders • To define key indicators and attributes to be monitored in the project.

The first phase of the project will set the baseline for its execution. The waste flows addressed by the project and the related upstream/downstream processes will be analysed assessing which data sets are missing or require new instrumentation/data exchange procedures for gathering accurate information and characterize each value chains and key players. The selection of information will be performed together with the definition of KPIs to be used for monitoring and assessment tasks. Waste manager partners will oversee the data collection, while RTOs will provide their support in data requirements and KPIs, and in the monitorisation and digitalisation of the processes. The consortium counts with the expertise of LTU as expert on digital innovation, who will set up the basis in this phase for a traceability system able to ensure the circularity of the involved value. Also, the governance structure to be implemented across the value chains and the participatory approach of REDOL will be defined to interact from the very beginning with citizens and key value chain entities, identifying main drivers and barriers for the project.

*The organisational, social and regulatory framework, characterisation of use cases, establishment of KPIs and development of the digitalisation requirements will be performed in **WP2 [M1-M15]**, setting the basis for REDOL*

P2 – Upgrading technologies for the management and processing of SUW.

Main Activities	• To carry out the detailed design of REDOL technical solutions. • To test and validate at TRL6 the prototypes. • To carry out the detailed engineering of the I-US solutions prior to their demonstration. • To fine-tune the demonstration campaign linked to each development.

The characterization steps carried out during the first phase of the methodology will provide the required inputs for the development of technologies in P2. This stage will include continuous communication between RTO, technology providers and facility managers to make sure that no deviations occur between the development phase and waste manager´s needs. Due to the nature of the proposed solutions, this phase has been divided into 2 different stages:

- **P2.1 - Development of specific sorting technologies for the management of targeted SUW streams**: 1) Smart urban bins for plastic collection at city level, 2) AI-based sorting system for MMPP based on spectroscopy techniques, 3) Integrated sorting system for CDW including HSI cameras and robotics arms, 4) Novel sorting system for textiles based on composition, texture and colour and 5) Mobile automated DL-based sorting system adaptable to different SUW.
- **P.2.2 – Development of processing technologies for the recycling of addressed value chains**: 1) green solvolysis of composite derived from CDW, plastic and packaging, 2) treatment and formulation of CDW for new purposes in the cement/concrete industry, 3) selective dissolution technologies for MMPP valorisation, 4) IL-based glycolysis for textiles recycling and 5) PCB valorisation using DES to extract precious metals.

***WPs 3 and 4** will oversee the development and validation of these solutions for SUW valorisation **[M9-M32]**. These activities will provide initial designs and layouts before the activities linked to demonstration start **[M25]**.*

P3 – Development of digital tools for the monitorization, management and optimization of REDOL value chains and their linked I-US interactions among key players

Main Activities	• To generate the digital twin of REDOL I-US interactions with forecasting capabilities. • To provide tools for ensuring traceability of resources and logistic optimization. • To provide the tools for assessing the life cycle impact of REDOL.

- To develop an ad-hoc and secure I-US Platform that integrates REDOL digital solutions.

During this phase, a complete development of the REDOL I-US Platform will be addressed considering aspects such as interoperability, security and data management. The necessary procedures will be used to integrate the developed tools and models, as well as the mechanisms that will allow the interoperability among the different value chains and involved stakeholders. The offered set of solutions will provide support to key players on the most adequate strategies for monitoring, managing and optimizing collaborative and symbiotic interactions.

*Digital tools development will be carried out in **WP5** for two years **[M9-M32]**, whereas their integration into the I-US Platform and fine-tunning with real demonstration data will last for the two final years of the project **[M24-M48]**.*

P4 – Overall demonstration of REDOL technical, organisational and social innovations through the redesign of the addressed SUW streams into circular and symbiotic value chains.

Main Activities

- To deploy and demonstrate REDOL technologies at TRL7.
- To validate the new value chains in terms of resource efficiency and cost-effectiveness.
- To establish cooperative links between key players and deliver feasible business models.
- To exploit the synergies among value chains and the maximise sustainability viewpoint.

In this phase, the technical developments performed at P2 will be implemented and demonstrated together with the organisational and economic arrangements defined in P1, thus ensuring that each value chain is able to reach a SRL 7 by the end of the project. P4 will interact as well with REDOL platform (P3) to test the exchange of data and optimization of REDOL processes, logistics and symbiotic interactions. Also, eco-design activities will take place in this phase to maximize the environmental benefits of the new circular value chains established by REDOL. Finally, along P4 the main task responsible partners (i.e., main RTD partners and waste manager of the value chain) will count with the support of CIRCE as coordinator and WP leader to monitor main KPIs and execute a techno-economic analysis, which will enable the quantification of REDOL impacts.

*P4 will be performed within **WP6**, partly in parallel with P3 activities, and will last until the project's end **[M25-M48]**, validating both the developed technologies and their synergistic collaborations.*

P5 – Overall impact analysis, social actions and replication strategy.

Main Activities

- To provide contributions to decision makers and relevant EU bodies.
- To share the benefits of REDOL with citizens through social innovation activities
- To engage key stakeholders and study business models for REDOL outcomes
- To develop IPR and exploitation strategies.
- To assess and initiate the replicability of the project developments.

The final stage will ensure that all the progress and developments that will be achieved during the project are perfectly compiled. In this vein, an impact assessment will be done to pave the way for exploitation and replication, that will take place in parallel and after the validation campaigns. To define these strategies, special attention will be paid to current barriers for innovation, as well as regulatory and policy frameworks, providing recommendations that will be shared with main related initiatives and bodies (e.g., CCRI and ECESP). Moreover, ad-hoc business models will be defined for the exploitation of the solutions, identifying new market segments and the required steps to bring I-US solutions to market. The overall conclusions will be gathered in a handbook for the development of hubs for circularity (D7.1) and will be used in the first replication phase at Prato, Amsterdam and Bornholm.

*Social awareness, exploitation and replication activities cover **WP7 and WP8**. They will last from the first year of the project **[M1-M48]** and are expected to continue in the future.*

Finally, alongside the methodology stages, C&D activities (with special attention in establishing synergies with ECoP) will use as inputs the outcomes obtained in each WP, engaging the maximum possible audience for the exploitation of results. A more detailed description of dissemin. & Exploitation strategy is included in Section 2.2.

1.2.8. Synergies with other national and international projects

REDOL relies on the lessons from previous EU and national projects addressing CE and U-IS strategies. Hence, it will establish links with these projects to find synergies and collaboration opportunities, exchanging information and providing outcomes. Some of these projects in which REDOL partners are involved are listed in **Table 2.**

Table 2. List of innovation projects related to REDOL.

PLASTICE	HE (2022-2026) Participants: CIRCE, RINA	SHAREBOX SECURE SHARING	H2020 (2015-2019) Participants: IRIS
New technologies to integrate plastic waste in the Circular Economy: REDOL will connect its value chains with PLASTICE demo-site in Zaragoza, closing the loop by introducing plastics recycling by-products in local markets.		**Secure Management Platform for Shared Process Resources:** REDOL will learn from the project experiences how the digital solutions can provide robust and reliable information in real-time to share resources effectively and confidently.	
SYMSITES	HE (2022-2026) Participants: ICL, AITEX	H4C EUROPE	HE (2022-2025) Participants: CIRCE, STRA, ICL

Industrial Urban symbiosis and its social, economic and environmental impact on different European regions: REDOL will create and implement innovative social actions based on the knowledge and results obtained on SYMSITES. Lessons learned in SYMSITES will be used as well in the monitoring and impact evaluation strategies.	**Building a European Community of Practice of Hubs for Circularity:** REDOL will benefit from the work on best practices of Industrial and Urban Symbiosis (I-US) as well as the ECoP network of stakeholders. H4C ECoP will also feed REDOL with sustainable business models for the deployment of services related to I-US.
CORALIS H2020 (2020-2024) Participants: CIRCE, ICCS, RINA, EEIP	SCALER H2020 (2017-2020) Participants: STRA
Creation Of new value chain Relations through novel Approaches facilitating Long-term Industrial Symbiosis: REDOL will adapt the multi-criteria evaluation framework for IS and governance structures of CORALIS to the I-US concept and key actors.	SCALER provides mechanisms to accelerate the journey towards efficient and quick implementation of industrial symbiosis in the European process industry. REDOL will use best-practices on action plans and adapted solutions for industrial stakeholders and communities.
MERLIN H2020 (2021-2024) Participants: ITE, IRIS	polynSPIRE H2020 (2018-2022) Participants: CIRCE, AIT, RINA
Increasing the quality and rate of MultilayER packaging recycLINg waste: REDOL will learn best-practices from MERLIN on technologies for recycling packaging (TRL5). Additionally, knowledge in closing the plastic value chain will be adapted for their use in REDOL.	**Demonstration of Innovative Technologies towards a more efficient and sustainable plastic recycling**: REDOL will implement experiences on plastic recycling from this project, extrapolating these approaches at I-US level.

Circ Lean Four REDOL partners; CIRCE, EEIP, ICCS and IRIS are member of CircLean, a network of stakeholders interested in fostering the future of industrial symbiosis in the EU.

1.2.9. Interdisciplinarity

REDOL is comprised of a **multi-disciplinary team from 12 EU countries** with expertise in disciplines, covering waste management, sorting and sensing, materials processing and recycling, eco-design, citizens and stakeholders' engagement, all of them with wide experience in national and international projects. This balanced group ensures an adequate development of the project, considering all the required domains, highlighting the following:

- **Engineering and technology.** REDOL will combine several fields of engineering (e.g., chemical recycling, materials science, spectroscopy, robotics, computer science, etc.) in order to develop the targeted technologies and their integration in novel value chains.
- **Environmental.** Aiming at managing waste in an environmentally sound manner and making use of the secondary materials they contain, REDOL will contribute to make Europe the first digitally led circular, climate-neutral and sustainable economy. Recommendations issued from REDOL results will lower down the barriers related to transitioning to a modern, resource-efficient and competitive economy.
- **Social sciences.** REDOL will apply approaches from business economics, stakeholders' engagement, education and policy studies (WP7) to understand the needs of main stakeholders and citizens and ensure the full and wide acceptance of REDOL solutions. REDOL project will provide knowledge for developing a more resilient, inclusive and democratic European society facilitating knowledge transfer, deployment and uptake of project achievements.

This **multidisciplinary approach** is achieved thanks to REDOL consortium, which is explained in detail in Section 3.2. This partnership structure provides the complementarity and critical mass to accomplish the project goals successfully in terms of timing, quality and costs, ensuring that no partner is missing for the aimed activities.

1.2.10. Gender dimension

In accordance with EU's Gender Equality Strategy 2020-202518 and SDG-5 (Achieve gender equality), REDOL has been conceived on a gender equality basis. REDOL partners are **committed to consider gender dimension in all research activities and foster gender equality**, being fully aware of the Gender Equality in Horizon Europe guidelines. During the project, a full implementation of several use cases aims at covering the need of the whole population, regardless sex, gender or age. No direct output of the project directly affects women or men. Additionally, REDOL will plan (T2.2) and carry out engagement, networking, co-creation and knowledge-transfer activities (T7.3 and T8.3) in order to foster the citizens participation in the creation of Zaragoza's Circular Hub. When designing and implementing such activities, **gender dimension will be considered as a primary aspect** to avoid disparity and to foster an inclusive and democratic European society.

REDOL will foster gender equality by **promoting gender balance** in the involved research staff. The project is mainly linked to engineering activities, which is still considered a male oriented domain. In this context, since the consortium construction, the gender parity participation has been considered and, as an example, the project counts on a highly skilled woman as project coordinator. Consequently, REDOL will contribute to reduce the gaps in the science, technology, engineering and math areas identified by the European Institute of Gender Equality.

1.2.11. Open science practices

REDOL partners are committed to fostering the collaborative work and on a systematic sharing of knowledge and tools in the research process. REDOL will address this through the application of open science practices:

- **Early and open sharing of research:** In the first stage of WP where technologies will be developed and implemented (WP3-WP6), REDOL partners will preregister the methodology to increase transparency about the purposes and methods of the study. This preregistration document will be posted on a public available website (Open Science Framework, Registered Reports) and, once the preregistered study has been conducted, a report of the study and its results will be submitted for publication together with access to the preregistration document.
- **Involvement of relevant knowledge actors, including citizens:** REDOL will engage the citizens and key local stakeholders through the information sessions, co-creation workshops, awareness raising campaigns, events to foster participation on the project topics and gather their main concerns in the area, providing insights for the development of REDOL overall solutions. Engagement and awareness creation through knowledge sharing will be key to support the general public in the transition from passive consumers towards active citizens.
- **Research output management and measures to ensure reproducibility of research outputs:** REDOL will ensure the reproducibility of the outputs by covering the three main research processes that reproducibility is based on: reproduction, replication and re-use. All these processes rest on the availability of data and methods from the original study. All the outputs of the project (data, reports, publications, etc.) will be available in Open Access repositories either through preregistrations or through publications in open repositories or journals
- **Open access to research outputs and participation in open peer-review:** A dedicated budget for open access scientific publications is reserved. Open access clauses will appear in the Consortium Agreement (CA) and the Grant Agreement (GA). For every publication, no later than 6 months after the publication is released, a machine-readable electronic copy of the published manuscript will be available on the website. Research data and bibliographic metadata needed to validate published results will be stored in the repository respective area. Finally, to make project knowledge publicly available, GEO will exploit EC open access portals and tools (e.g., OpenAIRE, Horizon the EU Research and Innovation Magazine, Research*EU Results Magazine, etc.).

1.2.12. Research data management

REDOL project will carry out data collection in a set of use-cases to assess the technologies and effectiveness of the proposed solutions in real life conditions. All the data made available by the use-cases will be protected and available for REDOL partners and key value chains actors involved through the establishment of the circular community. The necessity of publishing all the data generated in the project, if it does not affect exploitation of results or competitiveness, is aligned with ethic issues and protects the privacy of data. REDOL will make data "'as open as possible, as closed as necessary", also with the aim to encourage collaboration across disciplines. Most research data generated by the project will be made open & will be offered to the Open Research Data Pilot (OpenAIRE), in which REDOL has declared its intention to participate. Whenever possible and in accordance with the exploitation and IPR strategy, developed web services and data will be uploaded and shared via the EOSC.

As an Innovation Action project, REDOL will provide the first version of a complete Data Management Plan (DMP) as an early deliverable within the first 6 months of the project (included in D1.1). The DMP will detail what data the project will generate, whether and how it will be exploited or made accessible for verification and re-use, and how it will be curated and preserved. It is expected that the DMP increases research efficiency, saves time and resources, enhances data security and minimises the risk of data loss. DMP will take special care to make the data findable, accessible, interoperable and reusable according to FAIR guiding principles:

- **Types of data/research outputs:** Most relevant datasets to be generated in REDOL (reviewed in the DMP) are: (i) data on value chains characterization-WP2; (ii) data on waste streams (iii) data on sensing and sorting technologies-WP3, (iv) data and optimization of processes, logistics and I-US interactions, (v) data obtained in developing Zaragoza's Circular Hub strategy in WP7 and (vi) data on urban-industrial models.
- **Findability of data/research outputs:** A metadata list structure is proposed to be used in the DMP. By carefully recording the metadata elements, researchers will have a ready checklist for determining whether datasets can be made available on an open data basis or in EC supported repositories: OpenAIRE, ROAR, OpenDOAR and (DORAS).
- **Accessibility of data/research outputs:** The CA will lay out the rules of confidentiality, including a breakdown of the background IP and define its access rights and/or license agreements, as well as the generated datasets and results (foreground IP) that will be open or not. It will also provide the protocols for accessing the data. Non-open research data will be archived in REDOL internal platform, while open data will be archived on Zenodo platform. Open research data will be uploaded as soon as the quality of the data is verified for the REDOL Data Manager.
- **Interoperability of data/research outputs:** To make data generated in REDOL interoperable, the following file formats will be considered: Spreadsheets (e.g., Microsoft Excel); Comma Separated Values (CSV); Text Documents: (i.e., DOCX, ODF, or PDF); Presentation (i.e..ppt); Image formats (i.e. .jpg, .png), Audio formats (.wav, .mp3), Video formats (.avi,). Quantitative data analysis will be stored in SAV file format from which data can be extracted

using the open source spread Perl script. DMP will define in detail the data and metadata vocabularies, standards, or methodologies that all partners will follow to make project data interoperable.

● **Reusability of data/research outputs:** As a preliminary assessment, REDOL will use an open and free of charge platform (e.g., Zenodo) to archive open research data. In any case, the DMP will deep into the terms and procedures to license data to permit the widest re-use possible. Also, the CA will optimize the creation of networks, interaction, exchange, and access to internal information/technologies and maximize technology transfer.

2. IMPACT

2.1. Project´s pathway towards impact

REDOL project is fully aligned with the **HORIZON-CL4-2022-TWIN-TRANSITION-01-10** and will contribute to the current EU policy framework and goals. Indeed, the Horizon Europe program aims at developing Key Enabling Technologies and implementing R&I activities that will be able to address the major stakes of the 21st century: **the achievement of carbon neutrality**, for which digital, low-carbon and circular industry is one of the core aspects of the EU strategy. This is the purpose of Cluster 4, "*Digital, Industry and Space*" which fulfils key strategic orientations (KSO) for EU. In particular, the 1st Destination of this Cluster, *"Climate neutral, circular and digitised production (Twin-Transition)"*, will address the following impact: *"Global leadership in clean and climate-neutral industrial value chains, circular economy and climate-neutral digital systems and infrastructures (networks, data centres)"*. REDOL project is thus **fully aligned with the key objectives of this destination**, making Europe the first digitally led circular, climate-neutral and sustainable economy through the transformation of its mobility, energy, construction and production systems. This will be possible by the generation of a set of **Key Exploitable Results (KERs)**:

Table 3. Key Exploitable Results of REDOL project.

Management & Sorting	Recycling & processing	Sensing/digital technologies	Non-technical
KER1. Smart Urban Bins **KER2.** Identification and classification of MMPP **KER3.** Integrated system for sorting CDW **KER4.** Integrated system for sorting textiles **KER5.** Mobile sorting system for SUW including novel AI algorithms and adaptable robots.	**KER6.** Selective dissolution of MMPP **KER7.** Green solvolysis assisted with MW and US **KER8.** Formulations of recycled cement/concrete **KER9.** IL-based glycolysis for textiles **KER10.** Removal of elastane **KER11.** DES-based process for PCB valorisation **KER12.** Novel circular products derived from SUW	**KER13.** I-US platform **KER14.** In-line sensors for SUW monitoring and traceability **KER15.** GIS-based mapping system **KER16.** Logistics optimization tool **KER17.** Forecasting and life cycle impact tools **KER18.** Data sharing blockchain system	**KER19.** Handbook on building hubs of circularity **KER20.** Guidelines on eco-design and value chains redesign **KER21.** I-US business models **KER22.** Recommendations on regulation and EU roadmaps for I-US **KER23.** Circular community

The overall outcomes of the REDOL project will originate from the combination of these results, that will allow to tackle holistically the current challenges of establishing SUW circular flows as explained hereinafter.

2.1.1. Contribution of the project to the expected outcomes specified in the topic

Deploy the concept of Industrial-Urban Symbiosis (I-US) on a real scale demonstrator, making the flow of solid waste circular in process, manufacturing and/or construction industries;

REDOL will connect process (CEM, BRI, SAI), manufacturing (TEC, MOS, FIN) and engineering and construction industries (ACC, CAS, RINA) with regional representatives and local authorities (ZAR, UZ), and waste managers and related stakeholders (TAT, HUSA, ACT and additional RTO partners and technology providers) within an active and vibrant network to share knowledge and practices related to I-US of SUW value chains. Such an approach will end-up in the establishment of a hub of circularity at regional level around the city of Zaragoza, addressing a total of 5 key value chains for the accomplishment of the EU CE Action Plan.

Reduce 80 % (in weight or volume) solid waste generated in comparison to current state-of-the art, by re-using, valorising and transforming waste, by-products and side-streams into new/secondary resources of raw materials;

Due to the combination of more efficient collection and sorting systems, recycling processes and residue valorisation solutions, REDOL aims at transforming Zaragoza into a zero SUW residues generation in 2030. To this goal, we also count with the support of other ongoing initiatives in the region like Zaragoza Cero Residuos, R que erre and Mission 100 Climate Neutral Cities, with which REDOL will try to maximize synergies. The project will also produce valuable and specific information about the sustainable performance of the addressed value chains over their whole life cycle, which would be available to replicate in different and broader environments. Likewise, REDOL will promote education/training activities with the participation of citizens and local authorities to reduce waste production and changes in consumer habits, which is expected to have a very substantial impact on total waste generation. Lastly, REDOL strategy focus on fostering the creation of I-US community of enterprises will provide a novel approach to realize the above expectation.

Plan actions (e.g., awareness of circularity potential) to overcome non-technological barriers for exploitation (i.e. waste regulations, standardisation, confidentiality and compliance, ownership, fair sharing of benefits, acceptance of the concept);

REDOL counts with a specific task at regional level (T2.3) and work package at EU level (WP7) to assess current regulations and other non-technical barriers and to analyse the replication of the project results into other EU countries. The results of these activities include public guidelines and learning material (D7.1 and D7.2) on circular economy, I-US, business models and recommendations based on the detected barriers. This will help to shape a proper policy environment enabling the seamless implementation of SUW recycling technologies.

Develop knowledge sharing: know-how, advantages, challenges and recommendations on technological and non-technological aspects (e.g., job profile optimisation) with the European Community of Practice (ECoP) and other relevant bodies, disseminating the major innovation outcomes to support the implementation of I-US;

REDOL will deliver policy recommendations to align main outputs with upcoming legislation on CE and recommendations on standards and certifications to ensure the secure integration of I-US relationships. The project will also address this dimension by providing guidelines to waste managers and governance structures to key actors of the value chains for implementing REDOL solutions. The involvement of industrial partners in all the use cases will ensure the representativeness of this analysis, being the cornerstone for the creation of a more secure culture for recyclers companies, equipment manufactures and technology providers. In addition, the project will also seek for synergies with the ECoP as an effective and sustainable forum/platform connecting hubs for circularity and encouraging I-US towards building new circular value chains, that will be materialized in a specific task (T7.5).

Explore and illustrate replication potential in other regions (e.g., by setting up a network amongst waste associations to optimise flow of secondary raw materials);

REDOL will follow a full value chain approach and will identify needs and opportunities beyond the valorisation processes themselves. The aim is to provide a set of guidelines to support the definition of sound business models and the establishment of trusting relationships and to overcome regulatory and standardisation bottlenecks. These outcomes will be for the key further replication of the results through EU and beyond, so RTO and engineering companies within the consortium will make use of them to become promoters and enablers of REDOL valorisation routes at the end of the project. To support the replication of the results, the project will deliver a platform to monitor the overall sustainability performance based on KPIs calculated along the whole life cycle of the products, verifying the environmental footprint of the project solutions.

Implement actions to facilitate relations and to involve the local community actors (authorities, associations, civil society, relevant businesses, especially SMEs, educational organisations, etc.), e.g., exchanging knowledge, training, human capital, contributing to the optimisation of job profiles and sharing with the local educational establishments and with the ECoP;

REDOL will apply approaches from business economics, stakeholders' engagement, education and policy studies (namely in WP7 and WP8) to understand the needs of end-users and citizens, ensuring the full and wide acceptance of REDOL solutions. At regional level, this will result in the establishment of a Circular Community, the cornerstone to populate the value chains with additional actors and increase awareness among citizens and local initiatives. In addition, REDOL will provide knowledge for developing high-quality jobs and pursuing global leadership of Europe's in the CE strategies, which will be implemented in already existing training courses on CE performed by REDOL partners (like UZ, CEM, CIRCE) and other educational programmes at EU level (T7.3).

Implement a social innovation spin-off action involving one of the local community actors.

Based on established social innovation initiatives like "rqueerre"[16], Zaragoza's forest[17] or programs providing a unique learning experience in a dynamic and inclusive environment (chair of sustainability), [18] common mechanisms will be developed and implemented under the leadership of Zaragoza's city council and university (ZAR and UZ) for engaging citizens applying a design thinking approach. These activities will enable REDOL to create awareness, facilitate knowledge transfer, deployment and uptake of project achievements and promote the active participation of society for a better collection and sorting of residues and for the initiation of entrepenurial actions.

2.1.2. Contribution of the project to wider impacts specified in the destination

Accelerate the twin green and digital transition of the manufacturing and construction sectors;

REDOL will contribute to both creating a more sustainable waste management industry and increasing the digitalisation of the addressed value chains, by developing a set of innovative valorisation routes together with cross-cutting digital tools and technologies to increase the efficiency, impact and replication potential. The analysis of the EU landscape and potential for REDOL replication will enable the project to contribute in the following years to

[16] Promoted by UZ to disseminate case studies that are carrying out circular economy and eco-innovation practices and give visibility to the challenges that companies face; https://www.rqueerre.com/

[17] Promoted by ZAR to foster the implementation of environmental friendly projects while contributing to the generation of a 700.000 trees forest within Zaragoza city; https://www.zaragoza.es/sede/portal/medioambiente/elbosquedeloszaragozanos/

[18] https://catedracemex.unizar.es/

digitalise and reduce the environmental impact of manufacturing and construction sectors linked to the management of SUW and/or making use of the secondary raw materials produced from their valorisation.

Create a new green, flexible and digital way to build and produce goods. This will lead to sustainable, flexible, responsive and resilient factories and value chains, enabled by digitisation, AI, data sharing, advanced robotics and modularity. At the same time, it will help reduce CO_2 emissions and waste in these sectors, and enhance the durability, reparability and re-cycling of products/components. It will also ensure better and more efficient use of construction-generated data to sustain competitiveness and greening of the sector;

REDOL will significantly contribute to increase the recycling of products/components by the development of a full circular economy approach including guidelines to design processes and products that enable the creation of circular value chains from its initial conception. The incorporation of AI techniques and forecasting tools in a holistic way by means of REDOL I-US platform will enable the proliferation of flexible processes that are able to face variability of supply, thus resulting in more resilient and sustainable value chains.

Make the jobs of the humans working in the manufacturing and construction sectors more attractive and safer, and point the way to opportunities for upskilling;

A dedicated task (T7.3) will address Health and Safety issues from the project processes, ensuring their strict compliance with all the safety requirements, and will develop learning resources towards different profiles (graduated and professionals). Accordingly, REDOL will foster the creation of a critical mass by training the new generation of workers and re-skilling and up-skilling the current professionals.

Set out a credible pathway to contributing to climate neutral, circular and digitalised energy intensive industries;

REDOL will provide a deeper understanding of the main challenges stakeholders involved in the recycling value chains face, including energy intensive industries such as SAI, BRI and CEM. Their overcoming by implementing new technologies and solutions on I-US will result in a set of recommendations included in REDOL handbook for the creation of hubs for circularity that will support additional energy intensive industries in actively participating in the implementation of circular SUW value chains to increase the sustainability of their processes.

Increase productivity, innovation capacity, resilience, sustainability and global competitiveness of European energy intensive industries. This includes as many as possible new large hubs for circularity by 2025 (TRL 7 or above); developing sustainable ways for circular utilisation of waste streams and CO_2/CO streams; and electrifying industry to enable and foster a switch to a renewable energy system;

REDOL will contribute to increase the sustainability and resilience of the whole SUW value chain, including resource and energy intensive industries. They will be benefited by the increasing amount of waste material reintroduced into the production cycle, which will enable the creation of circular value chains for the targeted streams. Additionally, the technologies developed by REDOL will be designed to contribute to a lower environmental impact, introducing green solvents (e.g., DES, ILs and $scCO_2$) and electrified processes (e.g. MW-assisted pyrolysis), reducing the dependence from fossil fuels for its performance. This approach will result in the creation of a hub for circularity in Aragon and in the generation of best-practices and recommendations for the establishment of new hubs along Europe, maximizing the impact of the project.

Contribute to a substantial reduction of waste and CO_2 emissions, turning them into alternative feedstocks to replace fossil-based raw materials and decrease reliance on imports;

The core of the project activities is the valorisation of SUW to close the production loop; hence, the results from REDOL will directly contribute to this impact by optimizing resources use, keeping resources in the loop and reducing landfilling in the targeted use cases.

2.1.3. Measurable impact factors

REDOL will directly impact in the **transition of Zaragoza towards a zero residues city**, aiming at achieving this objective by 2040. Taking into account that the current recycling rate is around 50%, this would imply **over 140,000 tons/year being re-used, valorised or transformed into secondary raw materials instead of landfilled.**[19] It must be considered that potential increments in the number of residues generated within the city would further increase this impact, as residues generation is currently increasing by 2% per year, but efforts are placed in achieving a reduction in these figures. In order to support this objective and based on the project expected results, **REDOL partners have agreed on a strategy that relies on**:

- **More efficient and smarter collection systems**. They will enable the recovery of reusable waste streams and the reduction of generated mixed waste (currently representing 68% of the total share: 197 ktons). Mixed waste

[19] Information used for the benchmark analysis and impacts calculation has been developed by project partners in the 2021 report analysing I-US potential on SUW in Zaragoza (reference 8 of REDOL), whereas SUW generation and treatment data of Zaragoza come from Zaragoza City Council, and the "Plan Gestión Integral de Residuos de Aragon (GIRA) 2018–2020".

represents a barrier for the landfilling targets as its recycling rate is almost two times lower than separated waste (40% versus close to 70% recycling efficiency).

- **The implementation of novel sorting, classification, recycling and valorisation technologies**. REDOL will increase separation and recycling rates from mixed wastes, as they represent the most challenging stream in the city, although valorisation technologies are available for separated wastes as well.

Accordingly, along the project execution and after its finalization through the scale-up and replication activities to be implemented by its partners, the following objectives have been established to reach Zero landfilling by 2040:

Table 4. Targets of Zaragoza strategy to reach ZERO landfilling by 2040.

Year	2020	2025	2030	2035
Landfilling rate	50% 144.720 tons	25% 72.531 tons	15% 43.519 tons	10% 29.013 tons
Recycling rates for mix waste in Zaragoza	40% 81.479 tons	55% 108.507 tons	60% 118.371 tons	65% 128.235 tons

In particular, REDOL impact on the value chains addressed is expected to reach the following figures:

- **Packaging value chain**: The overall packaging recycling rate by the end of the project is expected to be up to 70%, which represents **landfilling savings of over 24 ktons** when compared to current benchmark (44% recycling rate for almost 100 ktons of waste packaging in Zaragoza).
- **Plastics value chain**: Due to the currently low recycling rate on plastics (20%), the objective has been set in 55%, thus resulting in up to **22 ktons of plastics being recycled**. In addition, the objective is to maximise synergies with on-going initiatives such as PLASTICE, whose reactors for depolymerization will be linked to REDOL to reach near zero plastic waste in overall.
- **CDW value chain**: REDOL will make possible to classify and recycle mixed waste at CAS facilities, which represents about 50% of the total managed residues, 40.000 tons. The valorisations routes proposed within **REDOL will transform 80% of these residues, i.e., 32 ktons, into valuable construction and cement secondary raw materials and products**.
- **Textiles value chain**: Only 14% is of textile waste produced in Zaragoza is currently recycled based on separated collection procedures. The implementation of REDOL approach after its demonstration in Prato is aimed at classifying and recycling the most challenges textiles, such as those including elastane or mixed fibres of PET and cotton, thus making possible the **full recycling of the 12 ktons textile waste of the city**.
- **WEEE value chain**: Currently WEEE counts as 100% recycled waste due to the specific collection and management processes. However, costs related to manual sorting procedures and valorisation processes prevent the reintroduction in the market of their many valuable materials. REDOL will tackle both barriers to **enable the circularity of the whole 315 tons of WEEE managed by TAT per year**.

The achievement of such ambitious goals and overall impact will be based on the performance of REDOL project, which will be quantified according to a set of impact factors (IF) to be monitored and supervised to ensure their fulfilment by the end of the project execution. They have been divided into a total of 9 IF (**Table 5**).

Table 5. REDOL Impact factors

Impact factor	Quantification and rationale	Scale	Significance
IF1. Validated technologies	-10 technologies will be demonstrated in REDOL for the sorting and valorisation of SUW in addition to solutions on digitalisation and new materials formulation.	-Outputs will be of interest for all EU waste managers and relevant RTO and equipment manufacturers.	-Efficiency of SUW sorting over 90%. -Valorisation technologies enabling a zero residues approach.
IF2. New circular products in the market	-12 circular end-products validated in terms of economic and technical feasibility within the project, including issues such as regulation and health and safety.	-High relevance for manufacturing industries and end-users aiming to increase their sustainability metrics	-Contribution to the achievement of a zero residues economy while ensuring security of supply for process industries
IF3. Creation of hubs for circularity applying I-US approaches	-Aragon will establish a hub for circularity based on Zaragoza's city. REDOL will also set the basis for the creation of additional hubs for circularity in 3 EU cities: Prato, Bornholm and Amsterdam.	-EU municipalities in the track to accomplish EU objectives on CE and key players on waste management and valorisation.	-Contribution to EU targets: increase the level of preparation for reuse and recycling of municipal waste to 55 % by 2025 (65 % by 2035).
IF4. Business models and organizational	-New business and organisational models ensuring the full integration	-EU Industrial clusters, waste managers, and	-The implementation of REDOL and the strategy

procedures adapted to I-US	of the upgraded SUW value chains into the existing local waste management and industrial schemes.	any public or private institution willing to be involved in I-US.	for zero landfilling are calculated to achieve savings of over 14 B€.
IF5. Enhanced logistics and traceability of SUW flows and secondary raw materials	-Novel technology for inline sensing and traceability together with logistics optimization algorithms allowing for better decisions on resources management will be delivered	-Transport companies and waste managers in charge of collection systems in different European cities	-Increase monitoring capabilities in 20% -Logistics: improvement of up to 90% with respect to the initial situation
IF6. Number of standardizations working groups	-The consortium plans to be involved in at least 5 standardization groups: e.g., CEN/TC 249/WG 11 Plastics recycling, CEN/TC 249/WG 24 Environmental aspects, CEN/TC 249/WG 5 Thermoplastic profiles for building applications.	-Links with main EU standardisation groups, sharing inputs with other initiatives such as SPIRE, ECoP, CCRI, ECESP.	-Facilitate REDOL replication, thus increasing European competitiveness, sustainability and security of supply.
IF7. Evidence-based support for EU policies/targets in CE	-REDOL non-technical KERs, e.g., roadmaps, guidance documents and policy recommendations, will be provided to over 50 local authorities and around 20 EU institutions.	-The main target of IF7 are policy makers all over Europe, whether at international, national, regional or local level.	-Unlock the potential of I-US across Europe, facilitating the initiation of new initiatives and hubs for circularity
IF8. Creation of benefits for society as a whole (sustainable and resilient jobs)	-REDOL will implement social innovation actions to create awareness. Besides, the potential of CE to generate a net employment is expected to increase to about 700,000 jobs in the EU by 2030.[20] Thus, REDOL will act as employment multiplier at short term, creating +120 high-qualified jobs in Aragon (counting both direct and indirect).	-The target is the general public, both at local level and through Europe.	-60% of the Aragonese population is aware of what CE corresponds to and the work carried out in REDOL -Upgrade local talent and human capital -Improved consumer acceptance (by 70%) of SUW-derived products
IF9. Reduction of the urban environmental impact	-In addition to the reduction of residues being disposed around the city, according to previously cited analysis in this section, the approach to reach a zero landfilling city would enable savings of 282 $kTCO_{2eq}$	-The target is the general public, both at local level and through Europe.	-Decrease on GHG emissions, lower need for strategic feedstocks, and overall improved living conditions at urban level

2.1.4. Potential barriers

An analysis of the macro-environmental factors that can affect positively or negatively to the results deployment in the market has been carried out using the **PESTLE** approach. Political (P), economic (E), social (S), technological (T), legal (L) and environmental (E) factors that can have a hampering impact on REDOL are identified, as well as mitigation measures to overcome the identified barriers (**Table 6**).

Table 6. REDOL PESTLE analysis.

Factors	Barriers	Mitigation measures
Political.	It is crucial to involve Member States at the local/regional level to adequately address the challenges and needs of SUW collection, classification and processing, since they have at their disposal measures to encourage the adoption of solutions. Lack of standardization and regulations related to waste management can hinder the deployment of REDOL technologies and the marketability of the generated circular products.	• Main directives (e.g., the Packaging Waste Directive 94/62/EC) will be considered in REDOL eco-design tasks and in the provision of recommendations to regulatory bodies. • Public commitment will be sought to increase awareness and share knowledge generated in REDOL to reduce implementation timelines. • Convergence through EU policy instruments.
Economic.	High costs associated to collection, sorting and processing of SUW, hamper the funding of recycling processes, as cost can be higher than the value of the	• Detailed techno-economic analysis will be performed, and novel I-US business models will be created with the objective of spreading this

[20] Employment and Social Developments in Europe 2019. Chapter 5: Towards a greener future: employment and social impacts of climate change Policies.

secondary raw materials or obtained products. So, payback periods are crucial in decision making process in I-US interactions. Although slightly higher payback values are accepted when talking about sustainability and security of supply, the sector has strict requirements. Lack of solid business models to finance REDOL solutions in the long term constitute a barrier for many waste managers.	knowledge and facilitate the implementation of profitable circular value chains in the EU. • REDOL will make available data on available resources, including price forecasting and traceability, which facilitates planning, O&M and eases decision-making on investments for the modernization of waste management plants.
Social. In a context of greater sensitivity about the impact of our current consumption patterns, it is necessary to shape individual behaviours to be greener over time. The limited information accessible to the general public about the possibilities and benefits related to improving the circularity of waste in urban environments is a key obstacle to this. On the other hand, the lack of personnel specialized in recycling technologies implies a barrier for the implementation of the developed technologies and procedures.	• REDOL will make relevant effort in disseminating its results and benefits and will establish trust-worthy relationships between citizens and waste managers players. • REDOL will develop job profile programmes to improve capabilities and technical knowledge on developed solutions. • Local ecosystems will be benefited by means of job creation, both direct and indirect.
Technological. The adoption of some REDOL solutions may involve relevant reconfigurations and adjustments in existing assets and procedures. Additional barriers can include lack of data on waste resources, uncertainties of novel technologies and need for facilitative infrastructures for the replication of solutions (size, instability, etc.).	• Digitalisation, modelling and the virtual assessment platform will provide detailed technical information to industrial partners. • REDOL will enable a better and cost-efficient management of the solutions, thus reducing the retrofitting impact of the value chains.
Legal. The Directive 2008/98/EC on waste framework defines the permits and registrations of waste treatment, and it is coupled with the Regulation (EC) No 1013/2006 which determines the procedures for end-of-waste criteria. Concerns regarding unknown normative and differences between national and local legislations can slow down consenting and licensing.	• Analysing regulatory barriers on SUW valorisation is an activity within REDOL that will result in ad-hoc recommendations. • Credibility created through operating early results and dissemination actions based on proven benefits will be the basis to foster positive changes in current legislation.
Environmental. No clear barriers are faced from the environmental perspective, as REDOL complies with the "do not harm" principle. However, REDOL managerial structure, public authorities and consumer engagement campaigns will detect any potential issues that could hamper achievement of sustainability**.**	• REDOL will demonstrate sustainable activities including resources savings, wastes avoidance and decrease of GHG emissions. • LCA, LCC and S-LCA tools will be integrated into the REDOL I-US Platform to ensure the sustainability of the defined business models.

2.2. Measures to maximise impact – Dissemination, exploitation and communication

To guarantee the uptake of REDOL results in the EU market, all partners are committed to maximise the potential impact of the project´s outputs in terms of its dissemination and communication actions to relevant stakeholders, including society, industry, technology providers, authorities, etc. Beyond these measures, the commercial exploitation of REDOL results will receive in-depth attention from the beginning of the project, resulting in the development of a business model and business plan for each KER and value chain of the project.

2.2.1. Dissemination actions

Dissemination actions start from the very beginning of the project through **WP8** to **identify and engage the most relevant stakeholders**. Furthermore, REDOL partners will strongly support the dissemination of the project results through their reputed communication channels and strong sector-related communities. During the project, the targeted groups will interiorise the project approach having access to the information from successful uses-cases from the value chains and synergies created between the circular community. In this context, **networking will be crucial** to exploit synergies with projects, initiatives such as the ECoP, professional associations and consumer organizations. By incorporating these interests on REDOL solutions it is expected that they will **support and upscale REDOL replication activities beyond the project lifetime.** To this end, REDOL has defined first outline of the dissemination strategy including **target audiences (WHO) and type of communication (WHAT):**

Table 7. REDOL main dissemination actions.

Civil society, general public, NGOs: Dissemination goals toward this group include: i) establishment and maintenance of adequate communication channels with all types of participants; ii) promotion of awareness about project goals and activities among the circular community; and iii) fostering active acceptance of the project activities and their impact; collecting necessary information; physical deployment of infrastructure and realization. **Stakeholders and target groups:** Citizens, neighbourhood clusters, consumer organizations, environmental associations, students & teachers, EU-Citizen.Science, ECOS, RREUSE, Future in Our Hands, Rethink Plastic
Waste managers: Key stakeholders for REDOL are waste management actors and recyclers, since they are the responsible of the waste streams that the project is targeting. Most of these streams are under valorised, leading to costs associated to their management with no economic return. Main messages will include: the feasibility to treat unsortable waste streams and to obtain valuable products from them, the possibility to increase the revenue sources for their companies, and the environmental and social benefits of reducing landfilling and incineration. **Stakeholders and target groups:** ChemRecEurope, EuRIC, FEAD, EURITS, CEWEP, singular private waste managers
Industry & economic stakeholders: The solutions developed by REDOL will stablish a harmonious co-existence between the citizens, waste managers and the local industry, as well as conclusions on the market analysis and roadmap of the developed solutions to show the potential market opportunities of implementing a circular community. **Stakeholders and target groups:** Factory managers, workers, bankers, insurers, investors, business developers
Policy makers and public bodies: Successful use cases of the value chains, lessons learnt in REDOL project, policy recommendations and means to standardise the use of urban solid waste as raw materials for industries. **Stakeholders and target groups:** EC, EU Parliament, City and regional authorities, EU policy and decision makers, innovation and policy advisors
Technical experts & scientific community: Conferences proceedings, publications in scientific and technological specialized magazines and platforms and peer-to-peer communication. The main messages to deliver are the technical results, innovation and progress beyond the SoA and future challenges emerged during the project execution. **Stakeholders and target groups:** ZWE, EPRO, PRE, EPRO, EERA, PRE, EPRC
Initiatives and projects: Synergies with project and initiatives both ongoing and finalised can save time, money and trouble when establishing fruitful knowledge exchange and at the same time enable a wider outreach of REDOL. **Stakeholders and target groups:** CircLean, CCRI, Circular Cities Declaration, Circular City Centre, European Associations, International Associations and initiatives, European Circular Economy Stakeholder Platform (ECESP).

Project key results

Table 8 summarises the main results of REDOL to be disseminated to the project audience:

Table 8. REDOL results for dissemination

Target groups	Key project results	Expected impact
Waste managers and value chain key actors	-D1.1. Project Handbook -D7.2. REDOL learning materials	-To train and build strong organisation culture for the implementation of REDOL actions
Technology providers and manufacturers	-D5.3. I-US REDOL platform -D3.2. Advances of REDOL automated sorting and classification technologies -D4.2. Advances of SUW valorisation technologies	-To foster the deployment of innovative solutions along waste management value chains of cities around Europe
Technical experts, scientific community and professional associations	-D1.1. Project Handbook -D4.2. Advances of REDOL SUW valorisation technologies -D6.2. Guidelines for the design of process and products towards circularity	-To promote the exchange of knowledge on technical results and innovations to accomplish better performances and overall solutions
Policy makers and public bodies	-D1.1. Project Handbook -D6.4. REDOL impact and catalogue of I-US solutions for the establishment of circular and symbiotic value chains -D7.1 Guidelines for Circular Hubs in Europe -D7.3. Roadmap for replicability and European recommendations	-To support the development of a suitable regulatory and standardisation landscape that supports the widespread and uptake of REDOL solutions, and to facilitate I-US interactions
Investors and business developers	-D8.3 Business models for the exploitation of REDOL value chains	I-ncrease I-US opportunities by means of increasing their BRL
General public	-Brochures and webpage contents, general communication. materials (news, magazine, etc.), REDOL legacy (videos, monographs).	-To increase awareness on re-using and recycling and to promote their participation in I-US initiatives

2.2.2. Exploitation of results

Aligned and coordinated with communication and dissemination, exploitation activities will aim at preparing the playground for the exploitation of the developed technological solutions maximizing their value. To this end, under the guidance of the Project Coordinator (PC) and aligned with the **Exploitation Manager (STRA),** the consortium will deal with **IPR protection** and with the definition of detailed **business models** and a subsequent **business plan**, which will form the basis for the consortium's exploitation strategy.

Business models and business plans: Building on REDOL business case

Business models are the main instrument to support the exploitation of project's results, so **WP8** will provide particular business models to each KER. In addition, **novel business models** based on CE and I-US will be created to support profitable, sustainable and long-lasting hubs for circularity. STRA and CIRCE expertise in this field will contribute to the creation of collaboration agreements and financial arrangements to establish shared business models. They will be based on the **overall business case for REDOL: the need for waste managers to increase their valorisation rates to reach a zero landfilling objective while exploiting synergies with both public entities and local industries**. At the same time, municipalities will be benefitted from better environmental conditions and monitoring capabilities, while process industries and producers will obtain a safe and European based supply of raw materials and from the transformation of their productive processes into green and sustainable ones.

It must be also highlighted that the presence and synergies of key technology developers and providers within REDOL consortium will enable the participants to **reach exploitation agreements** already **within the project framework** and carry out the **refinement of the initial solutions design**, so they are adequate for commercial production. Such an approach will enable REDOL to advance towards SRL7 on the business and management perspectives in the last phase of execution, thus facilitating further market uptake and commercialisation of solutions.

To draft a first business case at proposal level we describe 3 key elements: value proposition, engagement of customers/end users and revenue model.

- Value proposition. The optimization of the addressed value chains through CE approaches will supply benefits for all the involved sectors. This, combined with REDOL I-US framework, will provide a kit valuable solutions. With this perspective, the following elements will support the value proposition of REDOL: (i) reduction of environmental impacts to comply with environmental legislations, (ii) technical, economic and environmental feasibility in the application of the solutions, (iii) customization of the solutions to waste managers/industries depending on the real needs and the availability of resources, (iv) capability to provide recycled materials adapted to different sectors and needs, (v) risk reduction in the investment accomplishments and in their decision-making related activities, and (vi) improvement in competitiveness due to the better performance of equipment and industrial processes.
- Engagement of customers/end users. As main beneficiaries on REDOL solutions will be waste managers and industrial manufactures of goods. REDOL value chain map will provide the evaluation of new strategies, opening new business models, strategies and customer relationships for its main beneficiaries. Also, investors will be targeted as there is a new trend in financial companies providing support to implement CE measures in some industries, becoming an interesting partner to develop and exploit new business opportunities. For the exploitation of KERs, the project will take advantage from previous sale channels and contacts of consortium members, and commercial efforts will be highly increased with major actors working in the process, manufacturing and construction industries.
- Revenue streams. The most important aspect in this point for REDOL is the ability to offer an attractive payback period. In this sense, the payback periods expected for most of the solution is 4-5 years due to the on the rise recycling behaviour changes for industries, authorities and citizens. Some of the reasons why end users would like to implement REDOL solutions can be derived from getting a service to improve efficiencies, the importance of the environmental eco-labelling and increasingly significance of corporate social responsibility.

Exploitation strategy

1st phase → Focused on **investigating the exploitation opportunities** among the project partners, **developing an exploitation action plan** and **fine-tuning a list of key exploitable results (KER)**. STRA will first analyse each technology, compare to existing competitors, evaluated the market and marketability potential and will develop a general exploitation strategy (T8.4) for the most promising KER. All partners will participate in the exploitation meetings to elaborate an exploitation plan to ensure all identified results are covered, and an internal agreement on the selection of exploitable results and how they will be achieved exists. A policy and CA on IPR related issues will be also implemented. A first version of the exploitation plan will be presented as a result of the first exploitation meeting.

2nd phase → Aiming at the development of the exploitation strategy by answering crucial questions such as:

- **i)** ***WHAT*:** clarification of exploitation results and development of exploitation routes.
- **ii)** ***WHO:*** identification of lead partners to achieve each exploitable result. Identification of partners interested in exploiting their own results and those who could delegate this task to the exploitation partner
- **iii)** ***HOW:*** which form(s) will the exploitation of these results take (direct industrial use, patenting, licensing, technology transfer, publication, input to policy making, etc.).
- **iv)** ***BARRIERS:*** i.e., identification of (overlapping) IPR and commercialization-related wishes of each partner; conditions that need to be fulfilled to enable exploitation of the results - identification of risks for the achievement of exploitable results and actions to mitigate these risks; availability of knowledge and

resources among the participants and willingness to contribute to the results.

v) **IP issues** and ownership of knowledge, which is crucial and linked to the "who" aspect.

3rd phase → Meet market requirements and address potential user preferences. The consortium will make a pre-assessment of the results' market potential and analyse the competitive situation based on the *Porter Five Forces* approach. One task will be designed to get relevant information on the potential market size and the competition (T7.4). For each exploitable project result, the related partner(s) will develop suitable business models for commercialisation and/or replication and formulate a business plan using the CANVAS approach.

4th phase → Start-up creation and road to market. If the technology and partners are ready, it is possible to create a legal structure (new start-up), set up organizational procedures (IPR contracts) and team (job creation), to market the solution and signed the first commercial contracts before the end of the project.

Key exploitable results (KER)

Table 9. REDOL key exploitable results: Main partners responsible for their exploitation and IPR protection strategy

	Sorting					Recycling and processing							Digital						Non-technical				
KER	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Main responsible Partners	ICCS	IRIS	ICCS, CIRCE	NTT	CIRCE, RINA, AMB	ITE	AIT	VDZ, CAS, CEM	AITEX; BRI	NTT	RINA	TEC, ACC, CEM, ETN	CIRCE	LTU	CIRCE	LTU	CIRCE	SUIT	ICL, STRA, UZ	AIT	CIRCE, STRA	ICL, EEIP	CIRCE
P	X	X	X	X	X	X	X	X	X	X	X	X		X									
KH	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SL													X		X	X	X	X					
IS	X	X	X	X	X	X	X	X	X	X	X	X		X									
OA																			X	X	X	X	X

*P: patent; KH: know-how; SL: software licencing; IS: Industrial Secrecy; OA: open access

2.2.3. Communication actions

Communication activities will target the promotion of the project to various audiences, comprising the media and the general public, and raising awareness on the addressed topics and findings. These activities will be especially focused on: **i)** creating a project visual identity and public image; **ii)** providing up-to-date information about the project; **iii)** sustaining the diffusion of results to the general public; and **iv)** translating the scientific/technical results into messages for public outreach, comprehensible also by the non-technical audience. REDOL communication actions will ensure this information is shared with appropriate audiences using the most effective means and tools, at regional, national and international level. The interconnection of various communication tools, such as multimedia material, press media, audio-visual media, website and social media, will be used to achieve maximum impact (**Table 10**).

Table 10. REDOL main communication actions.

HOW (Tool/activity)	WHEN	Evaluation and target value
Project visual identity & captative logo, including QR code, common graphics and templates	Ready by M2	-Common public image/branding ensure visibility and recognition
Project website. Giving public access to relevant non-IP-sensitive results, items, audio-visual content, deliverables, etc. Private area for internal use.	Launched by M4	-Number of hits, country of provenance and online visits: 10,000 hits/year from 50 countries
Social media (Facebook, Twitter, LinkedIn, etc.) To increase outreach in partners existing online communities.	Launched by M1	-Nº of posts: 5,000 posts -Nº of members: 15,000 members
E-newsletters. To provide up-to-date information and support exchanges with the community. Website update.	From M4 (every 6 months)	-Nº of subscriptions to the service: +150 subscribers
Promotional material: leaflets, videos, posters, roll-up banner, factsheet, etc. Uploaded to the project website	M2-M6 and M40-M48	-Nº of brochure/leaflet: +2,500 -Video visualization. +1,500
Scientific and technical publications. Peer-review scientific journals and conferences proceedings Publishable version of deliverables in the project website. E.g., Sustainability, Journal of Cleaner Production, Resources conservation and recycling	Along with deliverables and achievement of results	-Nº of papers: +8 -Downloads of technical project publications: +70
Cooperation with other projects to forge communication, promote synergies and establish cluster collaborations	Project lifetime	-Collaboration established with EU related projects: +12
Social and professional networking with key sector associations, platforms and initiatives. E.g., CircLean, EU-Citizen.Science, CCRI or the Circular City Centre.	Project lifetime	-Collaboration agreements with relevant networks: +6

Press media contacts (TV, newspapers, radio, websites. etc.) either general or specialized at EU/international levels. A press kit will be developed containing identity, press release, general presentation, key messages, a list of FAQs and copyright free images.	On a regular basis throughout project execution	-Nº of press releases: +25 -Mail-outs & downloaded newsletters (per release): +500
Conferences and workshops. Includes project final conference covering future strategy, expectations and exploitation measures. E.g., World Congress on Waste Management & Recycling, Recycling Congress, Waste Management Europe Exhibition	From M6 (+1 per year)	-Nº of attended conferences: +8 -Nº of organised workshops: +2 -Nº participants Final Project Conference: +100
Any dissemination of results will duly display the EU emblem and include information on the EU funding.		

Considering that the main project's conclusions will be delivered at the very end of the tasks, REDOL communication/dissemination activities will remain after the completion of the project to show over a long period of time its final results. In this sense, the project website and its deliverables are envisaged to be active by CIRCE for 2 years after the finalization of the REDOL project, making also possible the gathering of further publications and scientific literature developed and communicated beyond the project lifetime.

2.2.4. Strategy and procedures for IP management

Strategy and procedures for IP management will be an integral part of the REDOL knowledge management and protection strategy, so **definition and management of the IPR** will be **registered in the REDOL CA**. To ensure the efficient management of IP and exploitation of the developed solutions **an exploitation-working group will be set up** consisting of one person per partner as contact point for IP and exploitation issues. The group will meet regularly to get inputs from all partners regarding any new result that could be commercialized and smooth its way to the market. This exploitation-working group will be managed by STRA whose main duty will be managing the operation of the group, keeping record of the generated results, coordinating the implementation of the IP strategy, integrating the inputs from all the partners, reviewing any communication disclosure of results that could harm IP and reporting to the project management. **IP management** will be developed considering the **basic principles** set in the *Annotated Model Grant Agreement*, as well as the *Consortium Agreement General Principles*, onto which this strategy and procedures are built. They can be summarized as follows (precise definitions will be agreed in CA):

Ownership of results: results are owned by the beneficiary that generates them, whereby '*results*' means any (tangible or intangible) output of the action, as well as any rights attached to it.

Joint ownership by several beneficiaries: whenever results have been jointly generated by two or more beneficiaries, and it is not possible to establish the respective contribution, such results shall be jointly. Joint owners must agree (in writing) on the allocation and terms of exercise of their joint ownership, signing a Joint Ownership Agreement (JOA) within 6 months from the generation of the result.

Agreement on background: Beneficiaries must identify and agree (in writing) on the background for the action. '*Background*' means any data, models, software solutions, know-how or information held by the beneficiaries before their accession to the GA and needed to implement the action or exploit the results.

Access rights: the beneficiaries have agreed on the principle of giving each other access – on a royalty-free basis – to the background needed to implement their own tasks under the action.

2.3. Summary

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
The main need REDOL will address is the lack of solutions and schemes that results in low recycling rates (45%) in cities across Europe. In particular, main needs are related to 1) lack of traceable data, generating low efficient collection schemes, 2) lack of effective sorting and classification mechanisms, resulting in high amounts of mixed waste, 3) lack of recycling and valorisation alternatives, hampering the creation of circular value chains, 4) lack of governance structures enabling public bodies, local industrial facilities and citizenship interaction in a way that novel schemes for I-US can be implemented.	**New collection, sorting and classifications systems (KER 1-5)**. To increase the efficiency of the waste resource separation processes and consequently upstream processes performance, innovations will be developed for each of the 5 value chains addressed by the project. **Novel recycling and valorisation technologies (KER 6-12)**. Novel technologies for avoiding landfilling include: Selective dissolution, green solvolysis assisted by US and MW, new materials formulations, glycolysis with ionic liquids, DES-based process for PCB and design of new circular products. **Innovative digitalisation solutions (KER 13-18)**. REDOL will produce new smart sensors for monitoring, using blockchain to make the information traceable. Modelling, mapping, forecasting and life cycle impact calculations tools based on this data will be generated and used within REDOL I-US platform for supporting the management and optimization of value chains. **Non-Technical results (KER 19-23)**. Recommendations, business models, guidelines and a handbook for the establishment of hubs for circularity will be shared with REDOL audience including its Circular Community.	***Dissemination:*** Scientific and technical publications, conferences, workshops, and events with local/regional authorities to identify barriers and setting the circular community (WP7&8). Networking based on the partners memberships in platforms and associations. ***Exploitation:*** The exploitation routes of the project results has been presented in Section 2.2.2 The preferable option will be patenting of the valorization technologies, as well as licensing of the software solutions. ***Communication:*** Logo and visual identity, online channels creation and updating, press media, project promotional material, cooperation with other projects and initiatives, networking, open field site showcases.

TARGET GROUPS	OUTCOMES	IMPACTS
Waste managers and recyclers, that will be able to valorise waste streams currently disposed to increase their revenue streams and turnover and minimise management costs **Process industries,** including both producers of alternative raw materials and their end-users to produce more sustainable products in a feasible and certifiable manner. **Technical and scientific community**: which will have access to the new technical knowledge and developments of the project to progress with the innovation in this field. **Policy makers, and public bodies,** that will be able to identify barriers and elaborate more effective policies to support circularity in the waste management value chains. **Society as a whole**, benefitting from reduced waste generation and GHG emissions.	**Implementation of 5 circular-by-design value chains**, involving all the necessary key actors together with the required organisational structure and business models, resulting in the **consolidation of Zaragoza's hub for circularity**. **Optimisation of novel technologies performance**, enabling efficiencies over 90% for sorting, between 55% and 100% for recycling and generating a total of 12 circular end-products. **Reduction of landfilled material.** Over 90 ktons in the value chains addressed and up to 140 ktons in total for Zaragoza until 2040, thus achieving the zero-landfilling target. **GHG emissions reduction** compared to the current benchmark in Zaragoza, achieving reduction of 282 ktCO_2 by 2040. **Increased economic benefit** by the valorisation of disposed waste, aiming at producing 14 B€ benefits until 2040. **Smart and digitalised plastic value chains,** increasing current monitoring capabilities in 20%. **Improved policies and standards** towards the implementation of I-US initiatives and circular waste management value chains.	***Scientific:*** New datasets and evidence about the technical feasibility of the circular value chains, including the required conditions for sorting and recycling technologies, influence of the new secondary raw materials and product quality and the environmental footprint of the new processes. Results will be included in reports, roadmaps and papers. ***Economical/technological:*** Creation of more sustainable, safe and competitive waste value chains due to the integration of greener and more efficient valorisation processes, smart and digital solutions, and the effective design of new products and processes supported by the I-US platform and the generation of adequate governance structures and business models. ***Societal:*** Reduction of waste-related pollution and GHG emissions and accessibility to a higher portfolio of recycled products. Creation of learning resources for workers upskilling and facilitation of the participation in new I-US initiatives.

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

3.1. Work plan and resources

In order to achieve the objectives established in Section 1.1, REDOL activities will be structured in 8 Work Packages (WPs). The overall scheme of the work plan is illustrated in **Figure 3** in form of a **Pert Chart**.

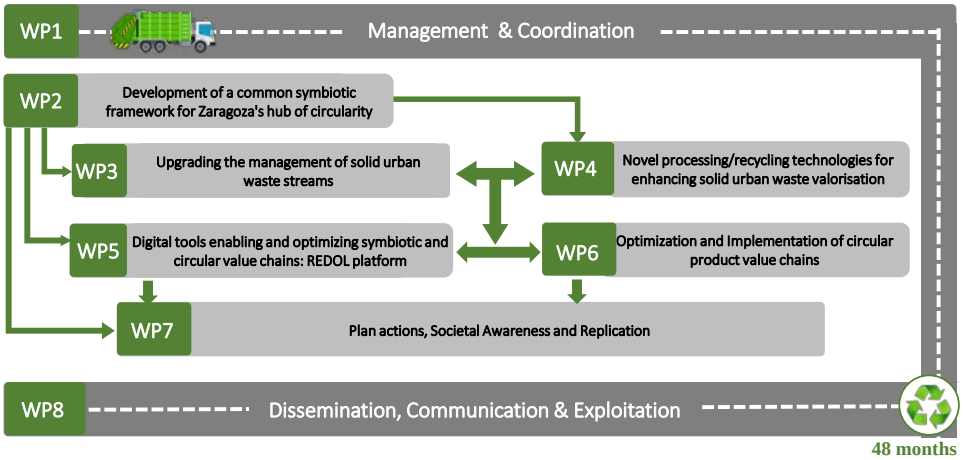

Figure 3. REDOL Pert chart

All the planned activities will be supported by coherent **management and coordination articulated in WP1**, which aims to develop an effective and comprehensive administrative, financial, technical and legal management. The technical activities will start with **WP2**, aiming to consolidate the project baseline and to obtain a deeper understanding on the preconditions that facilitate or hinder the implementation of I-US. This WP will make sure that a common understanding exists among all key players of the value chains regarding organisational, technical, regulatory, social and economic conditions that must be accomplished for the successful execution of REDOL.

Main developments of the project will take place in **WP3**, dealing with solutions on automated sorting and classification, and **WP4**, focussed on SUW valorisation technologies. These activities will rely on the detailed characterization of the value chains performed in WP2 and the close collaboration among waste managers, research entities and technology providers in order to provide with detailed designs validated at TRL5-6 for their final demonstration in REDOL value chains. Demonstration will therefore take place in **WP6**, where the value chains on packaging, plastics, CDW, textiles and WEEE will be redesigned to make them more circular. Moreover, a dedicated task has been included to address synergies among value chains, which will engage and interact with end-users of the valorised products to consolidate the new value chains.

REDOL results will be monitored and quantified according to a set of dedicated KPIs, and main conclusions will be gathered within **WP7** in order to ensure the replication of the project results. To this end, within this WP a handbook for the creation of circular hubs and key training material will be developed, and main regulatory, technical, economic and managerial recommendations will be gathered and shared with the ECoP and other relevant initiatives and stakeholders. These activities will be completed by a coherent exploitation strategy and dissemination plan to be developed and implemented within **WP8**. In addition to traditional outreach and exploitation activities, specific tasks have been devoted to engaging and interact with key stakeholders within the region for the set-up of the final value chains and to deploy and monitor circular and symbiotic business models.

A duration of **48 months** is foreseen to be sufficient to achieve all project objectives. As part of this methodology, a Gantt chart is provided (**page 30**), showing the timeframe of the activities planned.

Table 11. List of work packages

WP	Work Package Title	Partner	PMs	Start	End
1	Management and coordination	CIRCE	104.2	1	48
2	Development of a common symbiotic framework for Zaragoza's hub of circularity	AIT	206	1	15
3	Upgrading the management of solid urban waste streams	ICCS	258	9	32
4	Novel processing/recycling technologies for enhancing solid urban waste valorisation	ITE	319	9	32
5	Digital tools enabling and optimizing symbiotic and circular value chains: REDOL platform	CIRCE	191	9	48
6	Optimization and Implementation of circular product value chains	CIRCE	393.5	25	48
7	Plan actions, Societal Awareness and Replication	ICL	206.75	10	48
8	Dissemination, communication and exploitation	STRA	281.5	1	48
		Total Person Months	**1959.95**		

REDOL GANTT Chart

			Year 1												Year 2												Year 3												Year 4											
			M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	M19	M20	M21	M22	M23	M24	M25	M26	M27	M28	M29	M30	M31	M32	M33	M34	M35	M36	M37	M38	M39	M40	M41	M42	M43	M44	M45	M46	M47	M48
WP1	**CIRCE**	**Management and coordination**																																																
T1.1	CIRCE	Overall project management																																																
T1.2	CIRCE	Progress monitoring including reporting																																																
T1.3	CIRCE	Quality assurance plan and management and data management plan																																																
T1.4	CIRCE	Integration of the Responsible Research and Innovation approach																																																
WP2	**AITIIP**	**Development of a common symbiotic framework for Zaragoza's hub of circularity**																																																
T2.1	CIRCE	Implementation of the governance structure for U-IS interaction management																																																
T2.2	AITIIP	Establishment of the baseline for a participatory approach with citizens and stakeholders																																																
T2.3	UZ	Addressing non-technical barriers for the valorisation of SUW and their use as industrial feedstocks																																																
T2.4	AITIIP	Characterisation of value chains and key players interaction																																																
T2.5	UZ	Zaragoza waste resources mapping, logistics set up and digitalisation																																																
T2.6	CIRCE	Definition of the KPIs and impact evaluation methodology for the new circular value chains																																																
WP3	**ICCS**	**Upgrading the management of solid urban waste streams**																																																
T3.1	ICCS	Development and deployment of Smart Plastic Bins																																																
T3.2	IRIS	AI-based system for the identifaction and clasification of MMPP																																																
T3.3	ICCS	Integrated sorting system for CDW with advanced robotic arms																																																
T3.4	NTT	Novel textiles clasification solution using HSI camera																																																
T3.5	CIRCE	Mobile sorting system for SUW including novel AI algorithms, sensing technology and robots																																																
WP4	**ITENE**	**Novel processing/recycling technologies for enhancing solid urban waste valorization**																																																
T4.1	AITIIP	Valorisation of composite wastes through green solvolysis assisted by MW and US																																																
T4.2	ITENE	Selective delamination of MMPP recycling																																																
T4.3	VDZ	Formulation of recycled components for the construction and cement industry																																																
T4.4	AITEX	Recycling route for textiles based on IL-based, glycolysis, polymerization, post-condensation and spinning																																																
T4.5	RINA-CSM	Recovery of metals from WEEE-derived PCB using DES																																																
WP5	**CIRCE**	**Digital tools enabling and optimizing symbiotic and circular value chains: REDOL platform**																																																
T5.1	SUITE5	Interoperability for data sharing and confidentiality																																																
T5.2	CIRCE	Development of REDOL urban-industrial models																																																
T5.3	CIRCE	Applications for forecasting and pricing on waste resources																																																
T5.4	RINA-CSM	Algorithms for managing and optimizing logistics including trazability																																																
T5.5	CIRCE	Databases and tools for life cycle assessment and life cycle cost analysis																																																
T5.6	CIRCE	Development of the Industrial-Urban Simbiosis Virtual Assessment Platform																																																
WP6	**CIRCE**	**Optimization and Implementation of circular product value chains**																																																
T6.1	ITENE	Redesing of new circular packaging value chains																																																
T6.2	AITIIP	Redesing of new circular plastic value chains																																																
T6.3	ACCIONA	Redesing of new circular CDW value chains																																																
T6.4	AITEX	Redesing of new circular textiles value chains																																																
T6.5	CIRCE	Redesing of new circular WEEE value chains																																																
T6.6	CIRCE	Deployment of synergies across industrial sectors																																																
WP7	**ICLEI**	**Plan actions, Societal Awareness and Replication**																																																
T7.1	ICLEI	Development of the guidelines and strategy for building hubs of circularity																																																
T7.2	CIRCE	Analysis of regulations and barriers, and development of recommendations																																																
T7.3	UZ	Social innovation activities and Building human capital and job profile																																																
T7.4	EEIP	Understading the potential of U-IS around Europe: REDOL impact quantification																																																
T7.5	CIRCE	Interaction and contribution to major EU bodies on U-IS: ECoP																																																
T7.6	ICLEI	Roadmap for replicability of solid urban waste value chains across EU																																																
WP8	**STRANE**	**Dissemination, Communication and Exploitation**																																																
T8.1	GEO	Design of dissemination and communication strategy																																																
T8.2	GEO	Dissemination and communication actions																																																
T8.3	CIRCE	Stakeholder engagement and regional networking: REDOL Circular Community																																																
T8.4	STRANE	IPR review and exploitation strategy																																																
T8.5	STRANE	Assessment of business models for REDOL KER																																																
T8.6	CIRCE	Definition and implementation of U-IS business models within Zaragoza																																																

	Year 1												Year 2												Year 3												Year 4											
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
Kick-off General Assembly	X																																																
General Assembly													X												X												X											X
Steering Committee						X							X					X							X					X							X					X						X
Project Workshops and Final Conference	X												X												X												X											X

3.2. Capacity of participants and consortium as a whole

REDOL will be executed by a well-balanced and multi-disciplinary consortium composed by 35 partners from 12 European countries, particularly 5 large companies, 12 SMEs, 11 research organisations, 5 public institutions and 2 non-profit organisations. **CIRCE** will act as coordinator, its **wide experience in EU projects of relevance for REDOL** ensures the correct execution of the project. The consortium is a **high-quality partnership fully capable to attain the targeted objectives** and has been selected to include key actors essential to create synergies, ensuring that project results will maximize I-US symbiosis in the region and replication of the solutions across Europe. In this sense, **ZAR is the main public body** occupying the central hierarchical position of the hub for circularity and counts with the **signed commitment of waste managers such as URBASER and FCC** (currently in charge of waste collection and management in the city through public contracts) to support REDOL execution. The nexus with local industry and stakeholders is performed through **5 public and private waste managers and a total of 8 industrial facilities/end-users**. Their distribution along REDOL value chains is as follows:

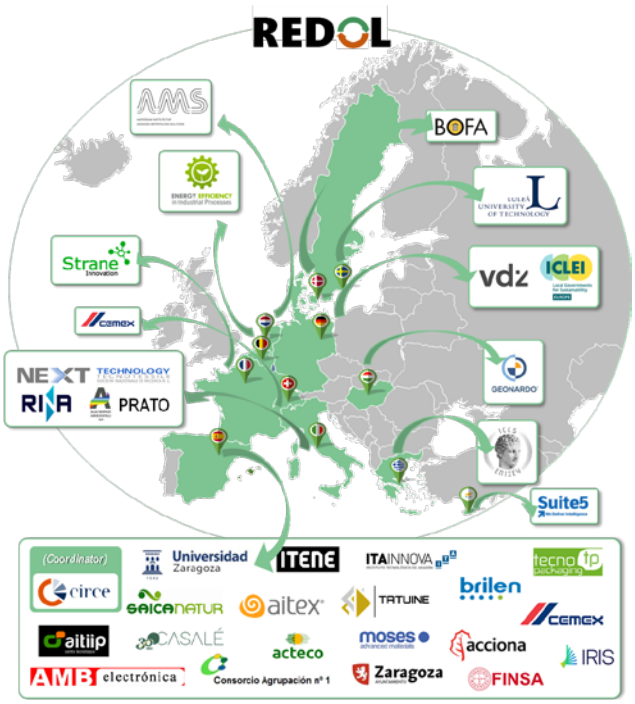

Figure 4: *REDOL geographical distribution*

Packaging	*Waste manager*	*RTO and Technology providers*	*Use Cases*	*End Users*
	HUSA	**ICCS, IRIS, ITE, AMB, CIRCE**	**SAI**	**ACC, TEC**
Plastic	*Waste manager*	*RTO and Technology providers*	*Use Cases*	*End-users*
	ZAR	**ICCS, AIT, CIRCE**	**ACT, MOS**	**TEC, CEM**
CDW	*Waste manager*	*RTO and Technology providers*	*Use Cases*	*End Users*
	CAS	**ICCS, VDZ**	**CEM**	**FIN, ACC, CEM**
Textiles	*Waste manager*	*RTO and Technology providers*	*Use Cases*	*End-Users*
	ALIA	**NTT, AITEX**	**BRI**	**ACC**
WEEE	*Waste manager*	*RTO and Technology providers*	*End users*	
	TAT	**CIRCE, RINA, AMB**	**ACC**	

Additionally, REDOL will be supported during the whole project by a set of horizontal partners with the objective of digitalising the whole value chains and waste management procedures, put in place the governance structure and business models needed for a successful I-US initiative, create a circular community and ensure citizens involvement in the transition to a CE by means of deploying social innovation and training actions. Together with these activities, coherent replication, exploitation, dissemination and communication strategies will be put in place by partners of high relevance in the field of hubs for circularity, I-US and the transition of cities towards a CE.

Horizontal activities			
Digitalization	*Circular community creation*	*Dissemination and exploitation*	*Replication*
CIRCE, SUITE5, LTU, RINA	**CIRCE, ZAR, ITA, UZ**	**GEO, STRA**	**PRA, ALIA, AMS, BOFA, ICL, EEIP**

GEO: will lead the dissemination activities in WP8 due to their expertise in public communication of European research achievements. To this end, GEO adopts an integrated impact-based communication and dissemination approach that supports awareness, social acceptance and uptake of innovation from public-funded research.

EEIP: is a non-profit organisation in charge of REDOL impact assessment to pave the way for replication and exploitation activities. They will pay special attention to current obstacles and barriers for innovation, as well as regulatory and policy frameworks for REDOL impact quantification.

STRA: will be the leader of WP8 due to their experience in exploitation activities and IPR reviews. They will also evaluated new ad-hoc business models, identifying new market segments and the required steps to bring REDOL solutions to market.

ICL: will lead WP7 developing best practices guidelines for the implementation Hubs of circularity from the analysis of demonstrators to accelerate the deployment and development of REDOL solutions.

To ensure a fluid coordination of the project REDOL will propose a coherent **management structure** for the activities monitoring and the decision-making processes. A tentative management structure is provided in **Figure 5.** The **General Assembly** is the highest-level decision-making body and will be composed by one representative of each project partner. The **Steering Committee**, consisting of the Coordinator and WPs leaders, executes the day-to-day management of the project, being the main execution body. Finally, REDOL Circular Community will be in direct contact with the project coordinator and will interact with REDOL consortium along the project life cycle according to T8.3 description. In any case, the management structure and the main bodies and their responsibilities, as well as the management processes will be agreed by all project partners and reflected within the CA prior to the project start.

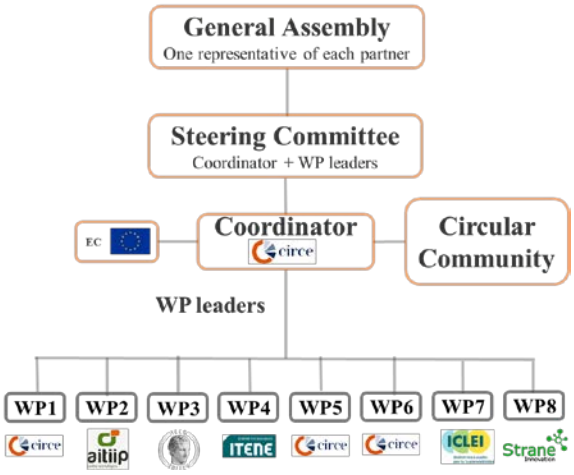

Figure 5. *REDOL management structure*

4. ETHICS SELF-ASSESMENT

Ethical dimension of the objectives, methodology and likely impact
REDOL objectives include an important ethical dimension related to its overall methodology and regarding the use of AI and advanced data analysis for the development of the sorting and classification systems (WP3). The AI systems developed in REDOL will guarantee privacy and data protection throughout their lifecycle. Data protection will be ensured by identifying individuals or groups to take on a specific role or task related to the IA system, in the first instance, and acquiring consent from all users whose data will be part of the algorithm system. Then, PLASTICE partners will use existing policies in dealing with user consent, use of and access to data, privacy, security practices, technology use, and human resource management to provide information about what policies may be missing and should be prioritized. Faced with the risk of using unsecured protected data, the mitigation strategy would be to have minimum data- and model-access requirements, including the prevention of sensitive data download. In the case of cybersecurity threats, real-time monitoring of the environment and maintenance of the capabilities will be ensured to respond rapidly to potential threats or vulnerabilities. All AI systems developed in REDOL will follow a transparency and accountability principle. The data sets used in the AI systems will be documented in English and all participants and end-users of the AI-based predictive tools will be trained on their proper use and informed about their limitations. The interaction with the AI systems, their abilities, limitations and benefits will be presented at meetings, and through the design of technical documents in English in order to explain their concepts to the end-users. The role of AI in the predictive tools developed will be made clear and explicit to the affected individuals by: (i) Building feedback requirements and feedback loops into the mode-development and redevelopment life cycle; (ii) Introducing the systematic tracking and reporting of all models performance including errors; (iii) Identifying clear roles and responsibilities for maintaining a view of regulations and their applicability to data management. Prevention of data bias will also be taken into consideration for the development of the AI systems to ensure optimum results are achieved. As each problem requires a different solution and provides varying data resources, there is no single model to follow that will avoid bias. However, there are several parameters that can inform as it is building: (i) To choose the right learning model for the problem; (ii) To choose a representative training data set; (iii) To monitor performance using real data. To minimize the risk of biased or discriminatory model outcomes, statistically significant input variables will be reviewed to validate their usability and the distribution of model results will be independently reviewed and validated to be free from bias.

Compliance with ethical principles and relevant legislations
REDOL will ensure respect for people and for human dignity and fair distribution of the benefits and burden of research, and will protect the values, rights and interests of the research participants. The necessary ethics approvals will be released before starting any activity and the free and fully informed consent of the research participants will be ensured. Regarding ethical needs for technology solutions to security and privacy, the Consortium will communicate and enforce participants data protection as a policy in all aspects of the approach with AI-based solutions, with the measures described above. Legislation such as the EU's General Data Protection Regulation (GDPR) (EU) 2016/679 for the control and protection of personal information provides the principles on which the handling of personal data will adhere to. Additionally, the wider understanding of privacy, such as those contained within the European Convention on Human Rights (Article 8), the EU Charter of Fundamental Rights (Articles 7 & 8) and related case law will also be considered and respected within the scope of the project, including definitions of privacy, and their relation to the security of information technology systems, as outlined within the ISO standard ISO/IEC 29134:2017. Also, legislation from the countries where the use cases are based (mainly Spain, but also Italy, Denmark and The Netherlands as replicators) will be followed and, in case of any differences with EU legislation, the strictest will be taken into consideration. REDOL methodology, including its AI systems, will be guided by ethical principles considering the "do not significant harm" (DNSH) principle as its activities do not significantly harm any of the six environmental objectives mentioned in Article 17, on the establishment of a framework to facilitate sustainable investment (EU Taxonomy Regulation). In fact, REDOL outcomes positively contribute to the objective "transition to a circular economy" by implementing technologies that enable the transition towards zero residues and circular value chains around SUW.